



Acute Effects of Cluster and Rest Redistribution Set Structures on Mechanical, Metabolic, and Perceptual Fatigue During and After Resistance Training: A Systematic Review and Meta-analysis

Ivan Jukic¹ · Amador García Ramos^{2,3} · Eric R. Helms¹ · Michael R. McGuigan¹ · James J. Tufano⁴

© Springer Nature Switzerland AG 2020

Abstract

Background The alteration of individual sets during resistance training (RT) is often used to allow for greater velocity and power outputs, reduce metabolite accumulation such as lactate and also reduce perceived exertion which can ultimately affect the resultant training adaptations. However, there are inconsistencies in the current body of evidence regarding the magnitude of the effects of alternative set structures (i.e., cluster sets and rest redistribution) on these acute mechanical, metabolic, and perceptual responses during and after RT.

Objective This study aimed to systematically review and meta-analyse current evidence on the differences between traditional and alternative (cluster and rest redistribution) set structures on acute mechanical, metabolic, and perceptual responses during and after RT, and to discuss potential reasons for the disparities noted in the literature.

Methods The Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines were followed, and five databases were searched until June 2019. Studies were included when they were written in English and compared at least one acute mechanical, metabolic, or perceptual response between traditional, cluster or traditional and rest redistribution set structures in healthy adults. Random-effects meta-analyses and meta-regressions were performed where possible.

Results Thirty-two studies were included. Pooled results revealed that alternative set structures allowed for greater absolute mean [standardized mean difference (SMD)=0.60] and peak velocity (SMD=0.41), and mean (SMD=0.33) and peak power (SMD=0.38) during RT. In addition, alternative set structures were also highly effective at mitigating a decline in velocity and power variables during (SMD=0.83–1.97) and after RT (SMD=0.58) as well as reducing lactate accumulation (SMD=1.61) and perceived exertion (SMD=0.81). These effects of alternative set structures on velocity and power decline and maintenance during RT were considerably larger than for absolute velocity and power variables. Sub-group analyses controlling for each alternative set structure independently showed that cluster sets were generally more effective than rest redistribution in alleviating mechanical, metabolic, and perceptual markers of fatigue.

Conclusion Alternative set structures can reduce mechanical fatigue, perceptual exertion, and metabolic stress during and after RT. However, fundamental differences in the amount of total rest time results in cluster sets generally being more effective than rest redistribution in alleviating fatigue-induced changes during RT, which highlights the importance of classifying them independently in research and in practice. Additionally, absolute values (i.e., mean session velocity or power), as well as decline and maintenance of the mechanical outcomes during RT, and residual mechanical fatigue after RT, are all affected differently by alternative set structures, suggesting that these variables may provide distinct information that can inform future training decisions.

Protocol Registration The original protocol was prospectively registered (CRD42019138954) with the PROSPERO (International Prospective Register of Systematic Reviews).

1 Introduction

Resistance training (RT) is traditionally utilized by athletes to increase their muscular strength, power, speed, endurance,

balance, coordination and lean body mass, which can ultimately lead to improvements in sport performance tasks [1, 2]. In addition, RT has an important role in injury prevention and rehabilitation, as well as general well-being due to numerous beneficial effects on health and quality of life [3–5]. To design an efficacious RT program tailored to the

Extended author information available on the last page of the article

Key Points

Cluster and rest redistribution set structures allow for greater absolute velocity and power outputs during resistance training, but these effects are even larger when all repetitions are expressed relative to the best repetition or when expressed as a decrease from the best repetition to the worst.

Cluster and rest redistribution set structures are effective in reducing the lactate accumulation and perceived exertion imposed by RT compared to traditional set structures.

Fundamental differences in total rest time result in cluster sets generally being more effective than rest redistribution for alleviating fatigue-induced changes in performance during and after RT, which highlights the importance of classifying and using them independently both in research and in practice.

individual's training goals, several factors such as the choice and order of the exercises, load, number of repetitions and sets, frequency, and length of designated rest periods are typically considered [3, 6–8]. However, the ability to alter the structure of individual sets during RT has also gained considerable attention from the sport science community, since set structure alteration can be used to manipulate mechanical [9–14], metabolic [9, 15–18] and perceptual [19–23] responses, potentially impacting the resultant training adaptations [24–26].

Traditionally, repetitions within a set are performed consecutively without rest between repetitions, and with a specified inter-set period before commencing subsequent sets. These so called “traditional sets” (TS) are often used in RT programs to target specific portions of the force–velocity spectrum [1, 27–29]. When doing so, athletes are often encouraged to perform concentric muscle actions as explosively as possible since lifting at maximal intended velocity may induce superior adaptations in strength and power compared to voluntarily lifting at a submaximal velocity [30–32]. However, regardless of the number of repetitions performed and the loads used, performing multiple repetitions in a consecutive fashion inevitably results in fatigue accumulation as evidenced by a decrease in movement velocity, increased perceived exertion, and metabolite accumulation, such as lactate [9, 12, 18, 23]. Since all these events represent adverse mechanical, perceptual, and metabolic responses which could then compromise certain training adaptations, alternative set structures such as cluster and rest redistribution set structures can be used to combat fatigue and performance decrements associated with TS.

The implementation of cluster sets (CS), where intra-set rest periods are added within a training set alongside standard inter-set rest periods, is an effective strategy to off-set mechanical [12, 14, 33], perceptual [18, 20] and metabolic [16, 18, 25, 34] fatigue often seen during TS. While the exact mechanisms through which CS ameliorate fatigue are not clearly established, it can be hypothesized that longer and more frequent rest periods during CS allow for greater maintenance of phosphocreatine stores, adenosine triphosphate resynthesis and increased metabolite clearance within the working muscles, resulting in enhanced recovery and performance maintenance [35–40]. However, extra rest periods during CS might not always be feasible since they extend total training time, which is not always possible to do in real-world settings [12]. In this regard, researchers have investigated other possible alternatives to CS such as redistributing the total rest time of TS structures, abbreviating inter-set rest and shifting it to include shorter but more frequent rest intervals [41]. This strategy, known as a rest redistribution set structure (RR), is more effective than TS for maintaining mechanical performance and mitigating fatigue—similarly to CS—but avoids the increased training time associated with CS [23, 42–44]. Collectively, although both CS and RR structures appear to be superior to TS in acute performance maintenance, they are divergent concepts. However, although these protocols have been used interchangeably in the literature (for a more detailed discussion of the evolution and terminology of alternative set structures, see review by Tufano et al. [41]), it is important to make a clear distinction between them as they could provide a different training stimulus.

For instance, Tufano et al. [12], Garcia Ramos et al. [9] and Gonzalez-Hernandez et al. [18] all showed CS to positively affect mechanical, metabolic and perceptual responses during RT. While studies examining the effectiveness of RR often come to similar conclusions, the magnitude of these effects is generally lower than those reported in studies comparing TS and CS. This is not surprising considering CS contain additional rest compared to TS and RR structures. However, inconsistencies in magnitudes of the effect exist even among studies that independently compared either CS or RR to TS. This variability in the results is likely caused by various confounding factors such as training to muscular failure, the number of repetitions performed, or the loads used, all of which could contribute to the different magnitude of effects observed between TS and CS or RR. For example, it is reasonable to assume that the benefits of CS and RR are exaggerated when TS are performed to failure and when the number of repetitions performed, and loads used are increased. Furthermore, although less explored, recent studies suggest that the effects of CS and RR on post-session mechanical, perceptual and metabolic fatigue might differ from the fatigue examined during the session (i.e.,

after each repetition or set). Considering the above, CS and RR likely impose different training stimuli and have different benefits which vary depending upon several factors which limits the conclusions that can be drawn from the existing literature.

A recent systematic review [45] compared the acute neuromuscular responses of RT sessions using alternative set structures (grouping CS and RR, together) and TS. The authors of the review concluded that alternative set structures allow for higher absolute velocity, power and peak force during RT, but also pointed out that several factors could potentially influence the observed differences (e.g., greater differences when using moderate-heavy loads). Although a valuable insight into the effectiveness of alternative set structures was provided, it is important to note that the specific, selective effects of CS and RR structures were not examined, and potential moderators such as the number of repetitions performed and muscular failure were not considered, limiting the generalizability of the findings. Furthermore, evidence regarding changes in mechanical performance during RT as well as mechanical, metabolic, and perceptual responses after RT differing only in set structure have not been synthesised to date. If we only observe the results from individual studies comparing either CS or RR to TS, the conclusions regarding the effects of CS and RR on mechanical, perceptual and metabolic responses might be that: (1) CS and RR are equally effective; (2) CS are more effective than RR; or (3) RR is more effective than CS. This clearly demonstrates the inconsistencies in the current body of evidence regarding the effects of CS and RR on the mechanical, perceptual and metabolic markers of fatigue during RT. Therefore, the aim of this systematic review and meta-analysis was to synthesise the available evidence on differences in mechanical, perceptual and metabolic responses between TS and CS or RR structures. This review will also provide an overview of the factors that may influence fatigue responses between the set structures, allowing researchers and practitioners to compare and contrast these set structures with more nuance. Such evidence is important to inform RT prescription strategies, ultimately allowing for better fatigue management.

2 Methods

2.1 Registration of Systematic Review Protocol

A systematic review of the literature was performed according to the *Cochrane Handbook for Systematic Reviews of Interventions* (version 5.1.0) and following the checklist for the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [46]. The original protocol was prospectively registered with the International

Prospective Register of Systematic Reviews (PROSPERO) in June 2019 and involved a larger systematic review project comparing TS, CS, and RR set structures during resistance training across a broader range of outcome measures (Registration number CRD42019138954). The protocol registration occurred before any of the searches were conducted.

2.2 Literature Search

A PICO strategy was used to build search criteria for electronic databases. It consisted of terms for traditional, cluster, and rest redistribution set structures; resistance training and acute mechanical, perceptual, and metabolic outcomes; and muscular strength, power and hypertrophy adaptations. The following bibliographic databases were searched from inception to June 6th, 2019: PubMed/MEDLINE, SCOPUS, Cumulative Index to Nursing and Allied Health (CINAHL), SPORTDiscus and Web of Science. The following syntax was adapted for each database and applied to the title, abstract, and keyword search fields: ("cluster set*" OR "rest redistribution" OR "set structure*" OR "set configuration*" OR "inter-set rest" OR "inter set rest" OR "intra-set rest" OR "intra set rest" OR "inter-repetition rest") AND ("strength" OR "resistance" OR "exercise" OR "training" OR "jump*" OR "plyo*" OR "weightlifting" OR "velocity" OR "power" OR "output" OR "velocity loss" OR "power loss" OR "velocity-loss" OR "power-loss" OR "force" OR "kinetic" OR "kinematic" OR "rpe" OR "rating of perceived exertion" OR "exertion" OR "effort" OR "perceptual" OR "lactic" OR "lactate" OR "fatigue" OR "metabolic" OR "hypertrophy" OR "muscle size" OR "muscle fiber" OR "growth" OR "cross-sectional area"). In the secondary search, the reference lists of all included publications were screened and the studies that cited the included studies were examined through the Google Scholar database. Although not relevant to this specific paper, muscular strength, power and hypertrophy terms were used since the search was a part of a larger systematic review project comparing traditional, cluster, and rest redistribution set structures. Free-text terms were chosen based on pilot searches that achieved an appropriate balance between search sensitivity and precision without the use of controlled vocabulary [e.g. medical subject headings (MeSH)]. The strategy included only terms relating to or describing the intervention. Conference abstracts and proceedings were excluded.

2.3 Inclusion Criteria

All studies included in the systematic review met the following inclusion criteria: (1) the study was peer-reviewed and published in English; (2) the study compared at least one acute mechanical, metabolic, or perceptual response between traditional and cluster or traditional and RR set structures;

(3) the study included a control group (i.e., TS) in which multiple repetitions within a set were performed in a consecutive fashion (i.e. without intra-set rest between repetitions) and from which intervention groups were derived (i.e. CS or RR); (4) the study included an intervention group (CS) which included intra-set rest intervals, resulting in longer total rest periods compared to the control group, and/or included a second intervention group (RR) during which total rest periods were the same compared to control group; (5) the load and volume (set $\times$ repetitions) were matched between the conditions; (6) the study included multiple sets of repetitions; (7) both intervention and control groups performed iso-inertial or bodyweight exercises; (8) participants had no known medical conditions or injuries.

2.4 Study Selection

Study eligibility assessments were completed independently by two reviewers (IJ and AGR). Records were downloaded into Endnote (version X8.2, Clarivate Analytics, Philadelphia, PA, USA) and duplicates were removed before being screened by title and abstract. The full texts of the remaining records were then retrieved and assessed for eligibility. Any discrepancies throughout the study selection process were settled through consultation with a third reviewer (JJT) or until a consensus decision was reached.

2.5 Study Coding and Data Extraction

From the studies that met the inclusion criteria, the following data were extracted into an Excel spreadsheet: (1) study identification information; (2) study design; (3) sample size; (4) participants' characteristics, including age, body mass, sex, strength levels, and training experience; (5) resistance training prescription details for TS, CS, and RR set structures; and (6) means and standard deviations for relevant outcome measures. If insufficient data were reported, the authors of those studies were contacted by-email. The Web Plot Digitizer software (Version 4.1; TX, USA) was used for the extraction of data from figures when the authors of the original studies did not report the data. Data extraction was completed independently by two authors (IJ and AGR) using a pilot-tested form on five randomly selected studies which was then modified accordingly. Coding files were cross-checked between the authors, and any observed differences were resolved via discussion and agreement, or with a third reviewer (JJT).

2.6 Risk of Bias Assessment

A risk of bias assessment was performed using a modified Cochrane Collaboration's tool for assessing the risk of bias in randomised trials [47]. Modifications included

removing the performance bias criterion and adding equipment bias, effort bias, and familiarisation bias. The performance bias criterion was removed because it was considered impossible to successfully blind participants and personnel in supervised exercise intervention studies [48–50]. Equipment bias was considered important for this review, and it was related to the reliability and appropriateness of the equipment or instruments that were used to assess outcomes of interest in a given study. For instance, if previous research failed to show the reliability of a given device being used in a study to record kinematic variables, this may have induced bias. Bias may have also arisen if an inappropriate instrument was used to measure the outcome (e.g. using rating of perceived exertion scales not validated for the use in resistance training). Effort bias related to whether the authors explicitly stated that all participants were constantly encouraged to perform the concentric phase of the movement as explosively as possible. Where this was not explicitly stated, there was a chance that the mechanical, metabolic, or perceptual measures of fatigue were affected by different tempo of lifting. Familiarization bias was related to whether the participants were adequately familiarised with the protocols in a given study. If the authors did not clearly state that a familiarisation session took place or that their participants were regularly performing exercises and performance tasks used in a given study, bias may have arisen from the participants being unaccustomed to the study protocols. This was deemed important regardless of the participants' training experience in a given study since performing exercises with maximal intent is not considered to be the norm and can highly influence the outcomes investigated in this review.

Risk of bias was assessed based on the information reported in the published papers, not on information provided in Table 1 of this review. Assessments were completed independently by two reviewers (IJ and AGR) while any observed differences were resolved via discussion and agreement before merging the scores into a single spreadsheet. Risk of bias was considered in the interpretation of the results by applying the Grading of Recommendations Assessment, Development and Evaluation (GRADE) system [51], similarly to previous reviews investigating physiological and biomechanical outcomes between different exercise modalities [48, 49]. Specifically, the overall quality of the evidence synthesis was rated as high and downgraded one level to moderate, low, or very low for each of the following limitations: total sample size < 100 participants (imprecision), high statistical heterogeneity (inconsistency), more than 50% of studies in a given meta-analysis had > 1 risk of bias item assessed to be high-risk (risk of bias).

Table 1 Summary of the studies included in the review

Study	Participants' information		Conditions				
	<i>N</i> (male); <i>N</i> (female); age (years);	Training experience; relative strength levels (BM/1RM)	Structure description Sets × repetitions; inter-set rest; intra-set rest	Load	Exercise	Total rest time	Outcomes
Davies et al. 2019 [71]	12; 9; 25 ± 7	Less than 1 year on average; 0.83 ± 0.14	TS: 4 × 5; 3 min; none CS: 4 × 5; 3 min; 30 s	85% 1RM	Free-weight bench press	TS: 15 min CS: 17 min	Mean velocity Peak velocity
Denton et al. 2006 [97]	10; 0; 25 ± 4	At least 1 year; unclear	TS: 4 × 6; 302 s; none RR: 8 × 3; 130 s; 1 min	85% 1RM	Smith-machine bench press	TS: 15 min and 10 s RR: 15 min and 10 s	Lactate
Garcia-Ramos et al. 2020 [9]	10; 0; 29 ± 4	3 years; 1.31 ± 0.21	TS: 3 × 10; 5 min; none CS1: 3 × 10; 5 min; 5 s CS2: 3 × 10; 5 min; 10 s CS3: 3 × 10; 5 min; 15 s	75% 1RM	Smith-machine bench press	TS: 10 min CS1: 12 min and 15 s CS2: 14 min and 30 s CS3: 16 min and 45 s	Mean velocity Lactate
Girman et al. 2014 [75]	11; 0; 23 ± 3	6 years; 1.76 ± unclear	TS1: 4 × 6; 2 min; none TS2: 2 × 4 + 2 × 10; 2 min; none RR1: 4 × (3 × 2); 1 min and 30 s; 15 s RR2: 2 × (2 × 2) + 2 × (5 × 2); matching the TS and RR; 15 s	50 – 75% 1RM	Free-weight clean pull and back squat	TS1: 8 min TS2: 10 min RR1: 8 min RR2: 10 min	Countermovement jump Standing long jump Lactate
González-Hernández et al. 2020 [18]	11; 0; 23 ± 2	Unclear	TS1: 3 × 10; 5 min; none TS2: 6 × 5; 5 min; none CS1: 3 × 10; 5 min; 10 s CS2: 3 × 10; 5 min; 15 s CS3: 3 × 10; 5 min; 30 s CS4: 3 × 10; none; 15 s	75% 1RM	Smith-machine back squat	TS1: 10 min TS2: 25 min CS1: 14 min and 30 s CS2: 16 min and 45 s CS3: 23 min and 30 s CS4: 7 min and 15 s	Mean velocity Countermovement jump Lactate RPE
Hansen et al. 2011 [65]	20; 0; 20 ± 2	Professional/Semi-professional rugby players; -	TS: 4 × 6; 3 min; none RR1: 4 × 6; 2 min; 12 s RR2: 4 × 6; 2 min; 30 s RR3: 4 × 6; 2 min; 50 s	40 kg absolute load	Free-weight jump squat	TS: 9 min RR1: 9 min RR2: 9 min RR3: 9 min	Peak velocity Peak power
Hardee et al. 2012 [11]	10; 0; 24 ± 1	4 years; 1.39 ± 0.03	TS: 3 × 6; 3 min; none CS1: 3 × 6; 3 min; 20 s CS2: 3 × 6; 3 min; 40 s	80% 1RM	Power clean	TS: 6 min CS1: 11 min CS2: 16 min	Peak velocity Peak power
Hardee et al. 2012 [20]	10; 0; 24 ± 1	4 years; 1.39 ± 0.03	TS: 3 × 6; 3 min; none CS1: 3 × 6; 3 min; 20 s CS2: 3 × 6; 3 min; 40 s	80% 1RM	Power clean	TS: 6 min CS1: 11 min CS2: 16 min	RPE

Table 1 (continued)

Study	Participants' information		Conditions				
	<i>N</i> (male); <i>N</i> (female); age (years);	Training experience; relative strength levels (BM/1RM)	Structure description Sets × repetitions; inter-set rest; intra-set rest	Load	Exercise	Total rest time	Outcomes
Hardee et al. 2013 [67]	10; 0; 24 ± 1	4 years; 1.39 ± 0.03	TS: 3 × 6; 3 min; none CS1: 3 × 6; 3 min; 20 s CS2: 3 × 6; 3 min; 40 s	80% 1RM	Power clean	TS: 6 min CS1: 11 min CS2: 16 min	Vertical displacement Horizontal displacement
Hoo et al. 2019 [72]	34; 0; 21 ± 2	unclear; ~ 1	TS: 3 × 6; 3 min; none RR1: 3 × 6; 2 min and 40 s; 20 s RR2: 3 × 6; 2 min and 20 s; 40 s	85% 1RM (6RM)	Free-weight bench press	TS: 6 min RR1: 6 min RR2: 6 min	RPE
Iglesias-Soler et al. 2012 [17]	10; 0; 23 ± 4	At least 18 months; 1.57 ± 0.34	TS: 3 × 4; 3 min; none RR: 1 × (360 s/number of repetitions completed in TS – 1); none; depended upon number of repetitions	90% 1RM (4RM)	Smith-machine back squat	TS: 6 min RR: 6 min	Mean propulsive velocity Lactate
Iglesias-Soler et al. 2014 [73]	9; 0; 24 ± 4	3 years; 1.57 ± 0.34	TS: 3 × 4; 3 min; none RR: 1 × (360 s/number of repetitions completed in TS – 1); none; depended upon number of repetitions	90% 1RM (4RM)	Smith-machine back squat	TS: 6 min RR: 6 min	Mean propulsive velocity
Joy et al. 2013 [98]	7; 0; 23 ± 2	At least 1 year; 1.77 ± unclear	TS: 4 × 10; 2 min; none RR: 4 × 10; 1 min; 1 min	75% 1RM	Free-weight back squat	TS: 9 min RR: 9 min	Mean power
Jukić et al. 2019 [19]	26; 0; 28 ± 5	At least 1 year; 1.82 ± 0.33	TS: 3 × 10; 4 min; none RR: 5 × 6; 2 min; none	70% 1RM	Free-weight back squat	TS: 8 min RR: 8 min	Mean velocity Mean power
Koefoed et al. 2018 [64]	10; 0; 27 ± 5	At least 1 year; unclear	TS: 4 × 6; 3 min; none RR: 4 × 6; 2 min and 20 s; 20 s	40% of Body Mass load	Free-weight jump squats	TS: 9 min RR: 9 min	Jump height Peak velocity Peak power
Kurt et al. 2018 [63]	13; 0; 21 ± 2	At least 1 year; unclear	TS: 2 × 10; 90 s; none RR1: 4 × 5; 30 s; none RR2: 10 × 2; 10 s; none	Body mass	Drop jumps	TS: 1 min and 30 s RR1: 1 min and 30 s RR2: 1 min and 30 s	Bounce drop jump height Reactive strength index Ground contact time

Table 1 (continued)

Study	Participants' information		Conditions				
	<i>N</i> (male); <i>N</i> (female); age (years);	Training experience; relative strength levels (BM/1RM)	Structure description Sets × repetitions; inter-set rest; intra-set rest	Load	Exercise	Total rest time	Outcomes
Mayo et al. 2014 [22]	7; 1; 24 ± 1	At least 6 months, unclear	TS: 5 × failure; 3 min; none RR: 1 × number of repetitions from a failure session (individualized); rest = (720 s / number of repetitions in failure session—1); depended upon number of repetitions	75% 1RM (10RM)	Smith-machine back squat Smith-machine bench press	TS: 12 min RR: 12 min	Mean propulsive velocity RPE
Mayo et al. 2019 [21]	12; 5; 23 ± 2	At least 6 months; unclear	TS: 5 × 8; 3 min; none RR1: 40 × 1 repetition; none; 18.5 s RR2 = 10 × 4; 1 min and 20 s; none	75% 1RM (6RM)	Leg press	TS: 12 min RR: 12 min	Mean velocity RPE
Merrigan et al. 2019 [43]	0; 12; 24 ± 4	At least 2 years; 1.6 ± 0.2	TS: 4 × 10; 2 min; none RR: 4 × 10; 1 min and 30 s; 30 s	70% 1RM	Free-weight back squat	TS: 9 min RR: 9 min	Mean velocity Mean power Peak velocity Peak power
Mora-Custodio et al. 2018 [16]	30; 0; 23 ± 3	At least 1 year and a half; 1.45 ± 0.24	TS: 3 × 6; 4 min; none CS1: 3 × 6; 4 min; 10 s CS2: 3 × 6; 4 min; 20 s	60% 1RM	Smith-machine back squat	TS: 8 min CS1: 10 min and 30 s CS2: 13 min	Mean propulsive velocity Countermovement jump Lactate
Morales-Artacho et al. 2018 [62]	18; 0; 24 ± 5	Unclear; 2.74 ± 0.41 (half squat)	TS: 6 × 6; 5 min; none RR: 6 × 6; 4 min; 30 s	20% Body Mass load	Countermovement jump	TS: 25 min RR: 25 min	Mean velocity Peak velocity
Moreno et al. 2014 [76]	26; 0; 22 ± 2	At least 6 months; unclear	TS: 2 × 10; 90 s; none RR1: 4 × 5; 30 s; none RR2: 10 × 2; 10 s; none	Body mass only	Countermovement jump	TS: 1 min and 30 s RR1: 1 min and 30 s RR2: 1 min and 30 s	Peak power Jump height
Nicholson et al. 2016 [25]	23; 0; 21 ± 2	At least 1 year; 1.6 ± 0.3	TS: 4 × 6; 5 min; none CS: 4 × 6; 5 min; 25 s	85% 1RM	Free-weight back squat	TS: 15 min CS: 23 min and 20 s	Mean velocity Mean power Lactate
Oliver et al. 2015 [70]	23; 0; 25 ± 5	Trained: at least 3 years; 1.75 ± 0.34 Untrained: unclear; 1.07 ± 0.38	TS: 4 × 10; 2 min; none RR: 4 × 10; 1 min and 30 s; 30 s	70% 1RM	Free-weight back squat	TS: 9 min RR: 9 min	Mean power Lactate
Oliver et al. 2016 [83]	24; 0; 25 ± 5	Trained: at least 3 years; 1.75 ± 0.34 Untrained: unclear; 1.07 ± 0.38	TS: 4 × 10; 2 min; none RR: 4 × 10; 1 min and 30 s; 30 s	70% 1RM	Free-weight back squat	TS: 9 min RR: 9 min	Mean velocity Mean power

Table 1 (continued)

Study	Participants' information		Conditions				
	<i>N</i> (male); <i>N</i> (female); age (years);	Training experience; relative strength levels (BM/1RM)	Structure description Sets × repetitions; inter-set rest; intra-set rest	Load	Exercise	Total rest time	Outcomes
Oliver et al. 2016 [34]	10; 0; 27 ± 4	At least 3 years; 1.96 ± 0.35	TS: 4 × 10; 2 min; none RR: 4 × 10; 1 min and 30 s; 30 s	70% 1RM	Free-weight back squat	TS: 9 min RR: 9 min	Mean velocity Mean power
Stone et al. 2019 [84]	8; 0; 24 ± 3	6 ± 3 years; 1.8 ± 0.3	TS: 4 × 6; 2 min; none RR: 4 × 6; 1 min and 30 s; 30 s	80% 1RM	Free-weight back squat	TS: 6 min RR: 6 min	Mean velocity Mean power
Torreon et al. 2019 [10]	13; 13; Male: 29 ± 2 years Female: 22 ± 1	Male: 6 ± 2 years; unclear Female: 1.2 ± 1 years; unclear	TS: 6 × 4; 3 min; none RR1: 6 × 4; 2 min and 45 s; 15 s RR2: 24 × 1; none; 39 s	85% 1RM (6RM)	Smith-machine back squat	TS: 15 min RR1: 15 min RR2: 15 min	Mean velocity
Tufano et al. 2016 [12]	12; 0; 26 ± 5	At least 6 months; 1.9 ± 0.23	TS: 3 × 12; 2 min; none CS1: 3 × 12; 2 min; 30 s CS2: 3 × 12; 2 min; 30 s	60% 1RM	Free-weight back squat	TS: 4 min CS1: 7 min CS2: 11 min and 30 s	Mean velocity Mean power
Wagle et al. 2018 [68]	11; 0; 26 ± 4	At least 1 year; 1.8 ± 0.3	TS: 3 × 5; 3 min; none CS: 3 × 5; 3 min; 30 s	80% 1RM	Free-weight back squat	TS: 6 min CS: 12 min	Mean velocity Peak velocity Peak power
Wagle et al. 2018 [74]	11; 0; 26 ± 4	At least 1 year; 1.8 ± 0.3	TS: 3 × 5; 3 min; none CS: 3 × 5; 3 min; 30 s	80% 1RM	Free-weight back squat	TS: 6 min CS: 12 min	Mean velocity Peak velocity Peak power
Wetmore et al. 2019 [69]	11; 0; 26 ± 4	At least 1 year; 1.8 ± 0.3	TS: 3 × 5; 3 min; none CS: 3 × 5; 3 min; 30 s	80% 1RM	Free-weight back squat	TS: 6 min CS: 12 min	Mean velocity Peak velocity Peak power

Values are expressed as means ± standard deviations

BM body mass, *CS* cluster set structure, *F* female, *M* male, *N*, number of participants, *RM* repetition maximum, *RPE* rating of perceived exertion, *RR* rest redistribution set structure, *TS* traditional sets

2.7 Statistical Analysis

A separate random-effects meta-analysis for each review outcome was performed using the *Meta* and *Metafor* statistical packages [52, 53] in R language and environment for statistical computing (version 3.6.3, The R foundation for Statistical Computing, Vienna, Austria) [54], when two or more studies reported on the same outcome. Model parameters were calculated by the restricted maximum likelihood method, and the observations were weighted by the inverse of the sampling variance [55]. For cross-over studies standardized mean differences with Hedges' *g* correction (SMD) and 95% CIs were calculated between traditional and alternative set structure trials based on the means and standard deviations of mechanical, perceptual, and metabolic outcomes, the correlation between the trials, and the number of participants. For between-subject studies, SMDs were calculated using the mean and pooled standard deviation data, as described by Borenstein et al. [56]. Since no studies reported any correlations between traditional and alternative set structures measurements, we estimated correlations when exact *p* values were available. However, it was often unclear whether studies reported post-hoc corrected or uncorrected *p* values, which resulted in implausible correlations. Therefore, to ensure that a maximal number of studies was included in the meta-analysis, a conservative correlation coefficient of 0.5 was used in all meta-analyses [57]. The SMD magnitude was interpreted as: small (0.20–0.49), moderate (0.50–0.79) and large (> 0.80) [58]. Sub-group analyses explored the effects of cluster and rest redistribution set structures on every outcome. Meta-regressions were performed when at least six effects were available for a given outcome [59] based on whether traditional sets were performed to muscular failure as a categorical covariate, and using the loads and the number of repetitions being performed in the traditional group as continuous covariates. The heterogeneity of the studies was assessed using the I^2 statistic. The I^2 statistic represents the percentage of total variation in estimated effects across studies due to heterogeneity rather than chance and was interpreted as small ($I^2 < 25\%$), moderate ($I^2 = 25\text{--}49\%$) and high ($I^2 > 50\%$) [60]. To explore the extent to which having both a within- and between-subject design in the same study impacted the pooled findings, we conducted a sensitivity analysis by excluding the studies with a between-subject design ($k = 4$) only for the outcome with the greatest number of studies. This was done since all studies with a between-subject design were included for that outcome (i.e., mean velocity) whereas only one or none of them were included for other outcomes in this review. Publication bias was not assessed because there was only a small number of studies

included in most meta-analyses and all studies were of a similar size, as recommended in the *Cochrane Handbook for Systematic Reviews of Interventions* [61]. In addition, there was no reason to expect that studies finding no difference between different set structures would be less likely to be published than studies reporting a statistically significant difference.

When multiple study effects were reported that were not of a direct interest to this review (e.g., separate effects for two or more TS, CS, or RR set structures) a combined effect was computed across these outcome measures with study-design specific formulas, as outlined by Borenstein et al. [56] for dependent and independent continuous outcomes to avoid double counting individuals from those studies in the meta-analysis. One study [10] reported separate effects for males and females. Since the sample size of each group was similar or higher than in most of the studies in the meta-analysis, the effects were not combined into one, but the groups were treated as two different studies to increase the statistical power. Similarly, Mora-Custodio et al. [16] reported separate effects for multiple loads and only the 60% 1RM load was used for the analysis since the other (greater) loads in study by Mora-Custodio et al. [16] were much more prevalent among the studies included in this review. This decision was further justified by the fact that the number of repetitions decreased as the load increased in the study by Mora-Custodio et al. [16] which resulted in extremely similar effects between the groups regardless of the load being studied. Four studies [62–65] in which countermovement or squat jumps were performed reported loads in absolute terms or percentages of participants' body mass. In this case, they were converted to percentages of back squat 1RM, based on findings by Cormie et al. [66], to simplify the analyses. When studies did not specify whether the repetitions during traditional sets were performed to muscular failure, the study protocol was classified as “to failure” when some of the participants either failed to complete the protocol or when the combination of the number of repetitions and the load clearly suggested that to be the case. When studies reported mean propulsive velocity instead of mean velocity, both variables were pooled and considered for the analysis. For the meta-analyses of mean velocity decrement, peak velocity decrement, mean power decrement, peak power decrement, velocity loss against a fixed load or jump height loss immediately after the session, lactate accumulation, and perceived exertion, effects were expressed such that positive values represent results in favour of CS or RR set structures over TS (i.e., lower decline in performance, lactate accumulation, and perceived effort).

3 Results

3.1 Search Results

The flow diagram of the literature search is presented in Fig. 1. The initial search from the databases resulted in a total of 1275 search results. After the removal of duplicates, the number of search results was reduced to 561. Title and

abstract screening resulted in the exclusion of an additional 498 search results. Of those 498, 63 full-text papers were screened based on inclusion/exclusion criteria. Forward citation tracking and reference list screening resulted in another 1434 publications to be screened, of which, four were included. Of those 67 papers, 35 full texts were excluded based on inclusion/exclusion criteria. As a result, the final number of studies included in this review is 32.

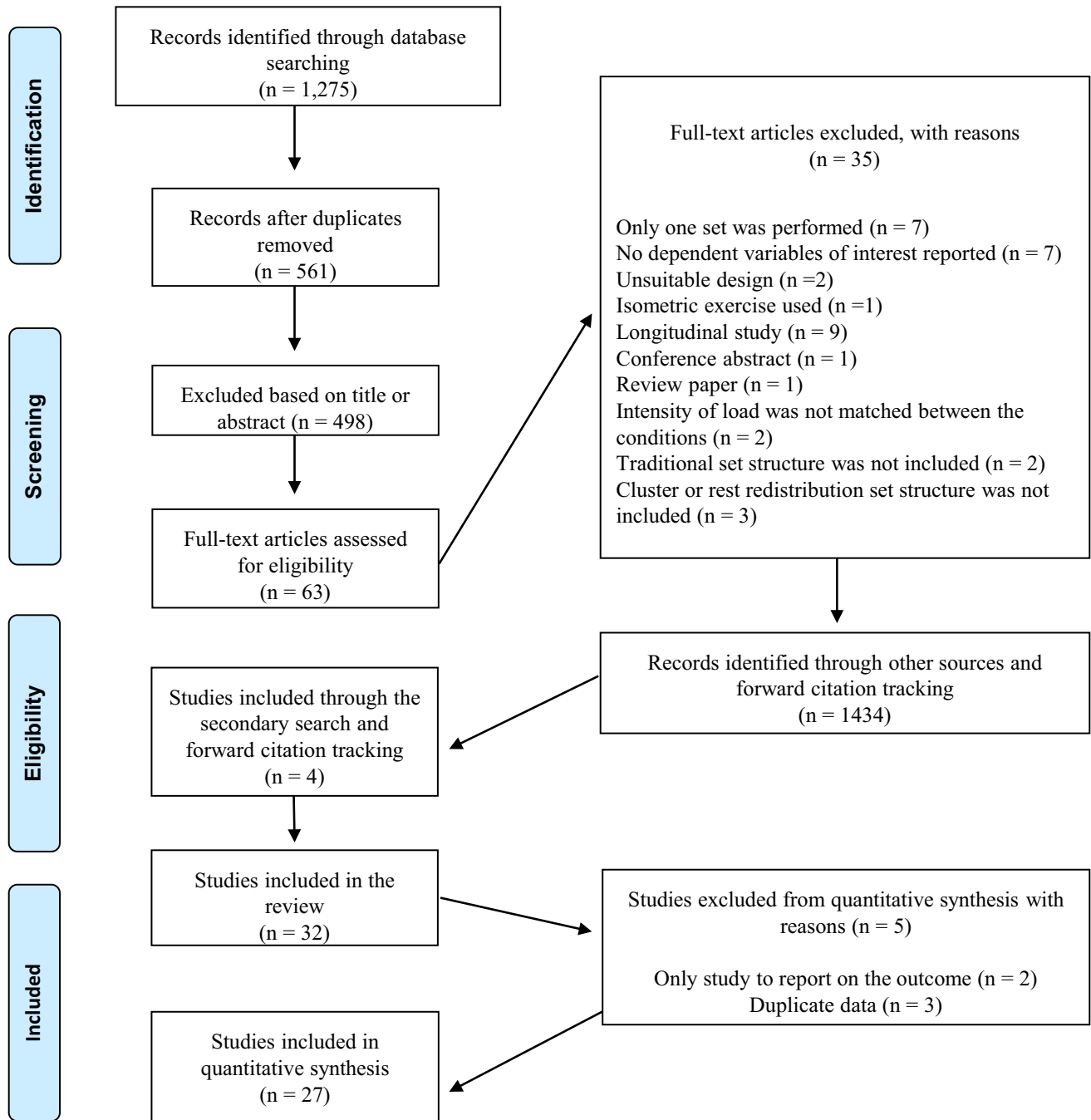


Fig. 1 Literature search flow chart. *n* number of studies

3.2 Study Characteristics

A total of 28 studies compared TS to alternative set structures with a within-subject design, and only 4 studies used a between-subject design. The pooled number of participants across studies was 481 (442 males and 39 females). However, after careful inspection, it was observed that data from the same participants were used in multiple studies [67–70]. Taking that into account, the total number of participants was reduced to 416 (377 males and 39 females). Of the studies included in this review, one study included females only [43], four studies included both males and females [10, 21, 22, 71], and the rest of the studies included only males. Twenty-three studies recruited participants with at least 1 year of resistance training experience, participants in 4 studies had at least 6 months of resistance training experience, and this information was unclear in 5 studies. Twenty studies compared RR to TS, and 12 studies compared CS to TS. In 11 studies, TS were performed to muscular failure. The median number of repetitions performed during TS in the included studies was 7 (range: 4–12) while the external load ranged from body mass to 90% of 1RM (median = 75% of 1RM). Twenty-three studies investigated free weight exercises, 8 studies used a Smith machine, and 1 study used a leg press machine. A more detailed description of the included studies can be found in Table 1.

3.3 Risk of Bias Assessment

Based on the risk of bias assessment, 5 studies [17, 22, 64, 72, 73] were at high risk of an order effect, not reporting any information about randomisation or having a fixed starting condition. Three studies [68, 69, 74] were classified to be of a low risk, while the rest were classified to be of an unclear risk since they did not provide sufficient information regarding the method of randomisation. No studies provided information regarding allocation concealment. One study [9] was constantly providing outcome feedback throughout experimental trials and therefore was considered to be of a high risk of outcome blinding bias. The remaining studies were rated to be at unclear risk of bias due to lack of information regarding the outcome blinding procedures. In addition, no studies pre-registered their protocols, so it was unclear whether the risks of selective reporting were present. One study [72] was at unclear risk of attrition bias, not reporting how the outcome of interest was collected when participants could not complete the protocols. Two studies were at a high risk of effort bias since the participants were not encouraged to perform the concentric phase of the lift with the maximal intent or they used a metronome instead [72, 75], whereas this information was unclear in one study [76]. Three studies were of a high risk of equipment bias since they used non-officially validated equipment to measure jump height

or movement velocity [19, 63, 75]. In addition, one study [72] used a rating of perceived exertion scale not validated for resistance training purposes and therefore, was also classified to be at a high risk of equipment bias. In total, eleven studies did not include a familiarisation session because of which a high risk of bias may have arisen due to the participants not being unaccustomed to the study protocols (Fig. 2). Three studies [11, 20, 67] were classified as of an unclear risk of familiarisation bias since it was not possible to judge whether the participants performed a familiarisation session or not. A more comprehensive risk of bias assessment is illustrated in Fig. 2.

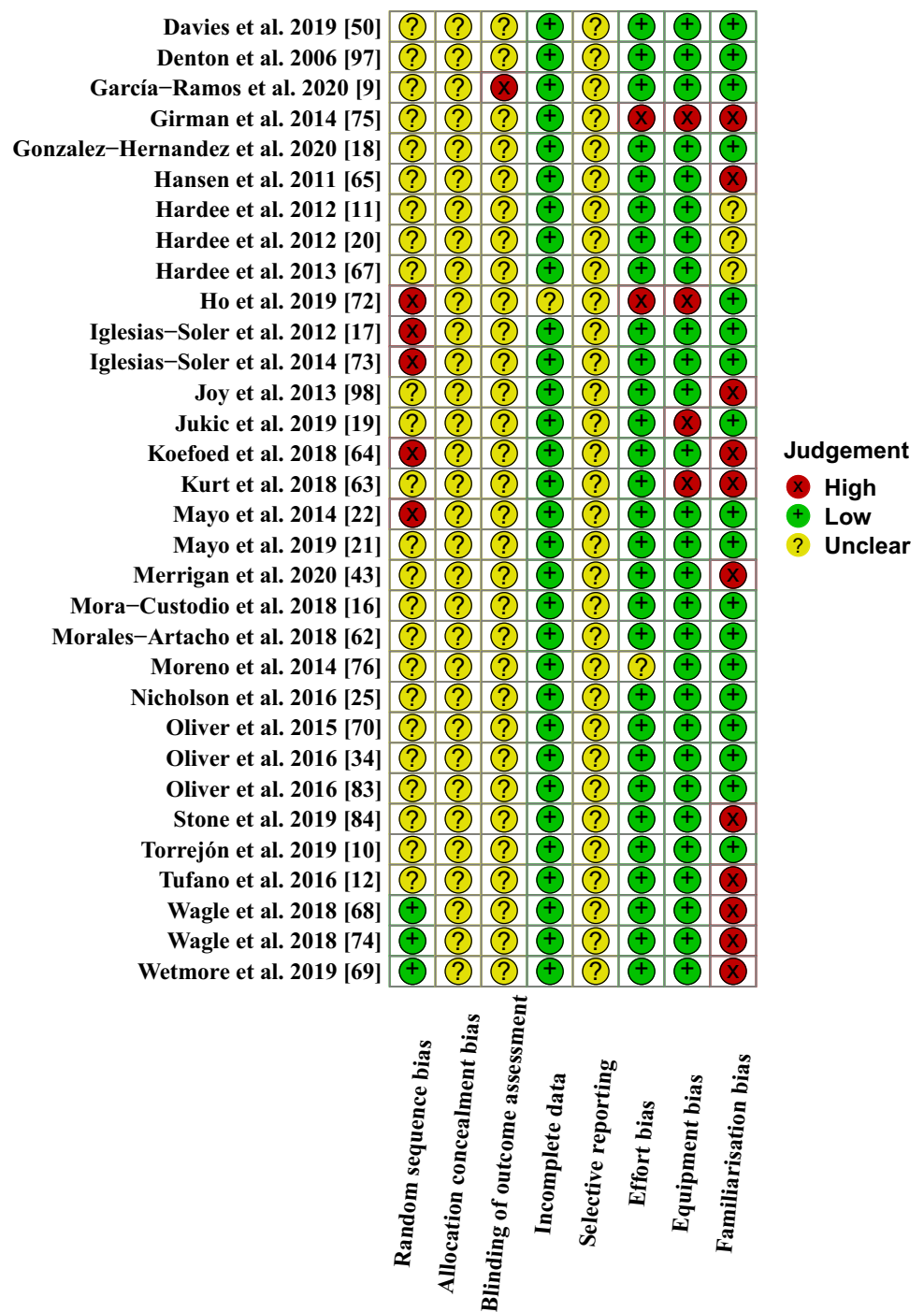
3.4 Mechanical Outcomes

3.4.1 Mean Velocity

Sub-group analysis indicated that both CS [SMD = 1.00 (0.72, 1.30); $p < 0.001$; $I^2 = 0\%$; $k = 7$] and RR [SMD = 0.39 (0.20, 0.58); $p < 0.001$; $I^2 = 33\%$; $k = 12$] set structures allowed for significantly greater mean velocity when compared to traditional set structures. When the effects from alternative set structures were pooled, mean velocity was significantly greater than during TS [SMD = 0.60 (0.40, 0.80); $p < 0.001$; $I^2 = 49.6\%$; $k = 19$] (Fig. 3). The sensitivity analysis, in which the four studies [16, 25, 62, 71] with a between-subject design were excluded, resulted in a pooled effect favouring alternative over TS [SMD = 0.55 (0.32, 0.78); $p < 0.001$; $I^2 = 55\%$; $k = 15$]. Pooled results and sub-group results for CS and RR all provided high GRADE quality of evidence (Table 2). Performing repetitions to muscular failure [$b = 0.15$ (– 0.27, 0.56); $p = 0.492$], the number of repetitions being performed during TS [$b = 0.01$ (– 0.07, 0.09); $p = 0.810$], and the load used [$b = -0.01$ (– 0.02, 0.01); $p = 0.456$] were not significant moderators of the SMD in mean velocity (Fig. 4). Likewise, when analysing CS and RR set structures separately, performing repetitions to muscular failure [CS: $b = 0.31$ (– 0.29, 0.92); $p = 0.307$; RR: $b = 0.22$ (– 0.15, 0.59); $p = 0.247$], the number of repetitions being performed during TS [CS: $b = 0.05$ (– 0.05, 0.15); $p = 0.322$; RR: $b = -0.03$ (– 0.10, 0.05); $p = 0.480$], and the load used [CS: $b = -0.01$ (– 0.03, 0.03); $p = 0.991$; RR: $b = -0.01$ (– 0.02, 0.01); $p = 0.468$] were not significant moderators of the SMD in mean velocity (Fig. 5).

3.4.2 Mean Power

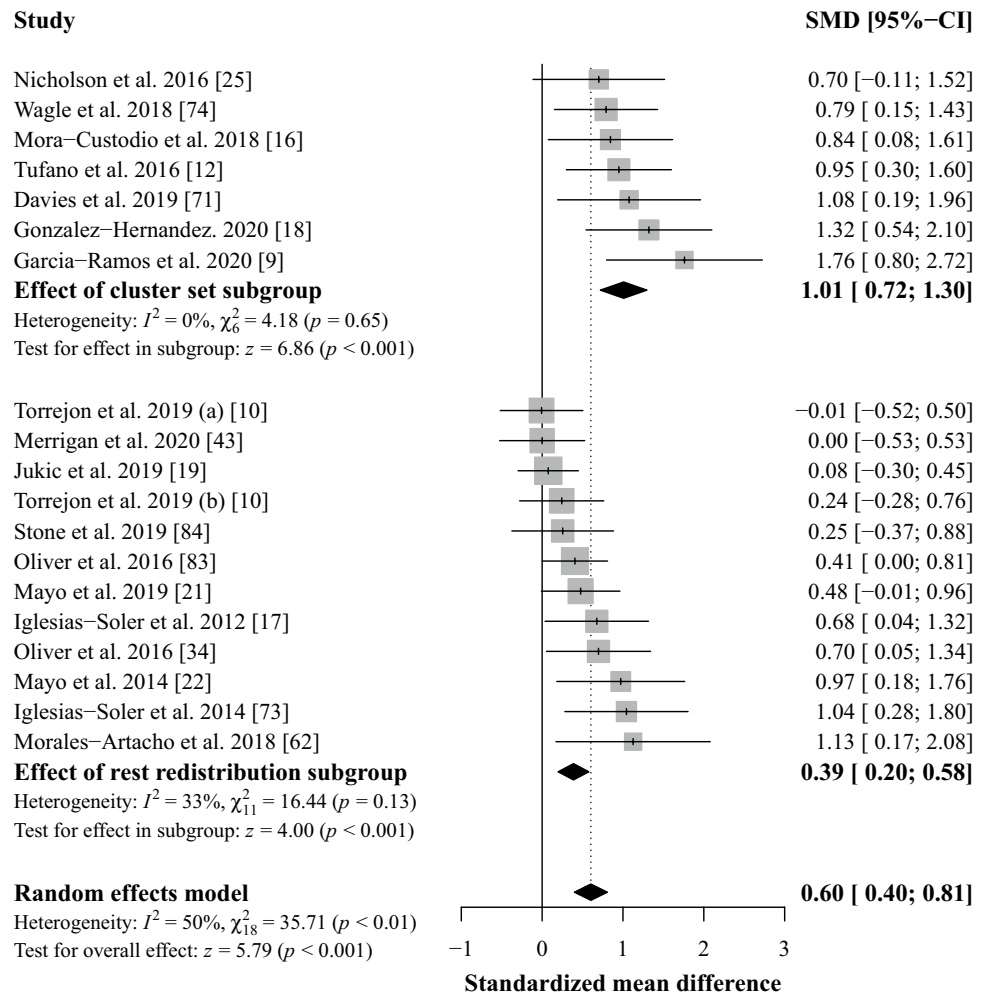
Subgroup analysis indicated that both CS [SMD = 0.83 (0.33, 1.34); $p = 0.001$; $I^2 = 0\%$; $k = 2$] and RR [SMD = 0.24 (0.04, 0.45); $p = 0.018$; $I^2 = 0\%$; $k = 6$] set structures allowed for significantly greater mean power when compared to TS (Fig. 6). When the effects from alternative set structures were pooled, mean power was significantly greater than

Fig. 2 Risk of bias assessment for all included studies

during TS [SMD = 0.33 (0.14, 0.51); $p < 0.001$; $I^2 = 0\%$; $k = 8$]. Sub-group results for CS and RR provided moderate while pooled results for provided high GRADE quality of evidence (Table 2). Performing repetitions to failure [$b = 0.19$ (− 0.20, 0.58) $p = 0.331$], the number of repetitions being performed during TS [$b = 0.03$ (− 0.09, 0.15); $p = 0.603$], and the load used [$b = -0.01$ (− 0.05, 0.02); $p = 0.509$] were not significant moderators of the SMD in

mean power (Fig. 7). Likewise, when analysing RR set structures separately, performing repetitions to muscular failure [$b = 0.25$ (− 0.17, 0.67); $p = 0.246$], the number of repetitions being performed during TS [$b = -0.01$ (− 0.08, 0.06); $p = 0.764$], and the load used [$b = -0.01$ (− 0.14, 0.19); $p = 0.300$] were not significant moderators of the SMD in mean velocity.

Fig. 3 Random-effects meta-analysis of absolute mean velocity during alternative set structures (i.e., CS and RR) compared to TS. Subgroup analysis based on the effects of CS and RR separately. For the study that analysed males and females separately, the effects are presented independently and are marked as (a) and (b), respectively. *CI* confidence interval, *CS* cluster set structure, *RR* rest redistribution set structure, *SMD* standardized mean difference, *TS* traditional set structure



3.4.3 Peak Velocity

Subgroup analysis indicated that both CS [SMD = 0.67 (0.29, 1.05); $p = 0.001$; $I^2 = 0\%$; $k = 3$] and RR [SMD = 0.28 (0.00, 0.55); $p = 0.050$; $I^2 = 0\%$; $k = 4$] set structures allowed for significantly greater peak velocity when compared to TS. When the effects from alternative set structures were pooled, peak velocity was significantly greater than during TS [SMD = 0.41 (0.19, 0.64); $p < 0.001$; $I^2 = 18\%$; $k = 7$]. Pooled results and sub-group results for CS and RR all provided moderate GRADE quality of evidence (Table 2). Neither the number of repetitions being performed during TS [$b = 0.01$ (-0.09, 0.11); $p = 0.847$] nor load used [$b = 0.003$ (-0.005, 0.01); $p = 0.404$] were significant moderators of the SMD in peak velocity.

3.4.4 Peak Power

Subgroup analysis indicated that CS [SMD = 0.54 (0.06, 1.03); $p = 0.027$; $I^2 = 26\%$; $k = 2$] but not RR [SMD = 0.30 (0.00, 0.55); $p = 0.050$; $I^2 = 0\%$; $k = 3$] set structures allowed

for significantly greater peak velocity when compared to TS. When the effects from alternative set structures were pooled, peak power was significantly greater than during TS [SMD = 0.38 (0.14, 0.63); $p = 0.002$; $I^2 = 4\%$; $k = 5$]. Pooled results and sub-group results for CS and RR all provided moderate GRADE quality of evidence (Table 2).

3.4.5 Mean Velocity Decrement

Subgroup analysis indicated that both CS [SMD = 2.32 (1.34, 3.30); $p < 0.001$; $I^2 = 71.5\%$; $k = 6$] and RR [SMD = 0.29 (0.04, 0.53); $p = 0.020$; $I^2 = 0\%$; $k = 4$] set structures resulted in significantly lower mean velocity decrement when compared to TS (Fig. 6). When the effects from alternative set structures were pooled, mean velocity decrement was significantly lower than during TS [SMD = 1.47 (0.63, 2.30); $p < 0.001$; $I^2 = 86\%$; $k = 10$]. Sub-group results for CS provided low while pooled results and sub-groups results for RR provided moderate GRADE quality of evidence (Table 2). The loads used [$b = -0.03$ (-0.12, 0.07); $p = 0.611$] or the number of repetitions performed during TS

Table 2 Summary of meta-analysis findings and quality of evidence synthesis

Outcome	Summary of Findings				Quality of evidence synthesis (GRADE)			
	k	n	Effect (95% CI)	Direction of the effect compared to traditional sets	Imprecision	Inconsistency	Risk of bias	Overall quality
Mechanical outcomes								
<i>Mean velocity (m/s)</i>								
Cluster set	7	118	1.01 (0.72, 1.30)	↑	None	None	None	High
Rest redistribution	12	151	0.39 (0.20, 0.58)	↑	None	None	None	High
All	19	269	0.60 (0.40, 0.81)	↑	None	None	None	High
<i>Mean power (W)</i>								
Cluster set	2	35	0.83 (0.33, 1.34)	↑	– 1	None	None	Moderate
Rest redistribution	6	80	0.24 (0.04, 0.45)	↑	– 1	None	None	Moderate
All	8	115	0.33 (0.14, 0.51)	↑	None	None	None	High
<i>Peak velocity (m/s)</i>								
Cluster set	3	44	0.67 (0.29, 1.05)	↑	– 1	None	None	Moderate
Rest redistribution	4	51	0.28 (0.00, 0.55)	↔	– 1	None	None	Moderate
All	7	95	0.41 (0.19, 0.64)	↑	– 1	None	None	Moderate
<i>Peak power (W)</i>								
Cluster set	2	23	0.54 (0.06, 1.03)	↑	– 1	None	None	Moderate
Rest redistribution	3	42	0.30 (– 0.01, 0.62)	↔	– 1	None	None	Moderate
All	5	65	0.38 (0.14, 0.63)	↑	– 1	None	None	Moderate
<i>Mean velocity decrement (%)</i>								
Cluster set	6	97	2.32 (1.34, 3.30)	↑	– 1	– 1	None	Low
Rest redistribution	4	64	0.29 (0.04, 0.53)	↑	– 1	None	None	Moderate
All	10	161	1.47 (0.63, 2.30)	↑	None	– 1	None	Moderate
<i>Mean velocity maintenance (%)</i>								
Cluster set	2	33	1.30 (0.70, 1.90)	↑	– 1	None	None	Moderate
Rest redistribution	2	38	0.62 (– 0.51, 1.75)	↔	– 1	– 1	None	Low
All	4	71	0.95 (0.24, 1.66)	↑	– 1	– 1	None	Low
<i>Mean power decrement (%)</i>								
Rest redistribution	3	46	0.70 (0.07, 1.33)	↑	– 1	– 1	None	Low
All	4	58	0.83 (0.28, 1.38)	↑	– 1	– 1	None	Low
<i>Mean power maintenance (%)</i>								
Rest redistribution	2	38	0.67 (– 0.56, 1.89)	↔	– 1	– 1	None	Low
All	3	50	0.78 (– 0.01, 1.57)	↔	– 1	– 1	None	Low
<i>Peak velocity decrement (%)</i>								
Cluster set	2	22	1.93 (0.14, 3.71)	↑	– 1	– 1	None	Low
Rest redistribution	2	32	2.08 (0.45, 3.70)	↑	– 1	– 1	None	Low
All	4	54	1.97 (0.98, 2.95)	↑	– 1	– 1	None	Low
<i>Peak velocity maintenance (%)</i>								
Cluster set	2	23	0.91 (0.39, 1.44)	↑	– 1	None	None	Moderate
All	3	35	1.36 (0.33, 2.40)	↑	– 1	– 1	None	Low
<i>Peak power decrement (%)</i>								
Cluster set	3	33	1.48 (0.82, 2.13)	↑	– 1	None	None	Moderate
Rest redistribution	3	58	1.95 (0.28, 3.62)	↑	– 1	– 1	None	Low

Table 2 (continued)

Outcome	Summary of Findings				Quality of evidence synthesis (GRADE)			
	k	n	Effect (95% CI)	Direction of the effect compared to traditional sets	Imprecision	Inconsistency	Risk of bias	Overall quality
All	6	91	1.71 (0.92, 2.49)	↑	– 1	– 1	None	Low
<i>Peak power maintenance (%)</i>								
All	2	24	1.47 (0.40, 2.55)	↑	– 1	– 1	None	Low
<i>Post session jump height loss or velocity against fixed load loss (%)</i>								
Cluster set	3	52	0.65 (0.29, 1.01)	↑	– 1	None	None	Moderate
All	4	62	0.58 (0.27, 0.89)	↑	– 1	None	None	Moderate
<i>Metabolic outcome</i>								
<i>Lactate (mmol·L⁻¹)</i>								
Cluster set	5	83	2.17 (1.18, 3.17)	↑	– 1	– 1	None	Low
Rest redistribution	3	43	1.27 (0.87, 1.66)	↑	– 1	None	None	Moderate
All	8	126	1.61 (1.18, 2.04)	↑	None	– 1	None	Moderate
<i>Perceptual outcome</i>								
<i>Rating of perceived exertion (a.u.)</i>								
Cluster set	2	21	1.14 (0.61, 1.67)	↑	– 1	None	None	Moderate
Rest redistribution	4	75	0.69 (0.28, 1.11)	↑	– 1	– 1	None	Low
All	6	96	0.81 (0.45, 1.17)	↑	– 1	– 1	None	Low

Only outcomes with $k > 1$ are included in this table

CI confidence intervals, GRADE Grading of Recommendations Assessment, Development and Evaluation, k number of studies, n number of participants

[$b = 0.13$ (– 0.19, 0.45); $p = 0.418$] were not significant moderators of the SMD for mean velocity decrement. However, performing repetitions to failure during traditional sets was a significant moderator of the SMD in mean velocity decrement with failure resulting in a greater difference in favour of alternative set structures [$b = 2.14$ (0.58, 3.70); $p = 0.007$] (Fig. 7). When analysing CS set structures separately, however, performing repetitions to muscular failure [$b = 1.44$ (– 0.69, 3.56); $p = 0.184$], the number of repetitions being performed during TS [$b = 0.01$ (– 0.12, 0.15); $p = 0.832$], and the load used [$b = 0.13$ (– 0.36, 0.62); $p = 0.594$] were not significant moderators of the SMD in mean velocity.

3.4.6 Mean Velocity Maintenance

Subgroup analysis indicated that CS [SMD = 1.30 (0.70, 1.90); $p < 0.001$; $I^2 = 11\%$; $k = 2$], but not RR [SMD = 0.62 (– 0.51, 1.75); $p = 0.283$; $I^2 = 87\%$; $k = 2$] set structures resulted in significantly greater mean velocity maintenance when compared to TS. When the effects from alternative set structures were pooled, mean velocity maintenance was significantly greater than during TS [SMD = 0.95 (0.24,

1.75); $p = 0.009$; $I^2 = 83\%$; $k = 4$]. Sub-group results for CS provided moderate whereas pooled results and sub-group results for RR provided low GRADE quality of evidence (Table 2).

3.4.7 Mean Power Decrement

Subgroup analysis indicated that both CS [SMD = 1.26 (0.53, 1.99); $p < 0.001$; $I^2 = 0\%$; $k = 1$] and RR [SMD = 0.70 (0.07, 1.33); $p = 0.029$; $I^2 = 70\%$; $k = 3$] set structures resulted in significantly lower mean power decrement when compared to TS. When the effects from alternative set structures were pooled, mean power decrement was significantly lower than during TS [SMD = 0.83 (0.28, 1.37); $p = 0.003$; $I^2 = 71\%$; $k = 4$]. GRADE quality of evidence for pooled results and sub-group results for RR was low (Table 2).

3.4.8 Mean Power Maintenance

Subgroup analysis indicated that CS [SMD = 1.09 (0.40, 1.77); $p = 0.002$; $I^2 = 0\%$; $k = 1$], but not RR [SMD = 0.67 (– 0.56, 1.89); $p = 0.286$; $I^2 = 88\%$; $k = 2$] set structures

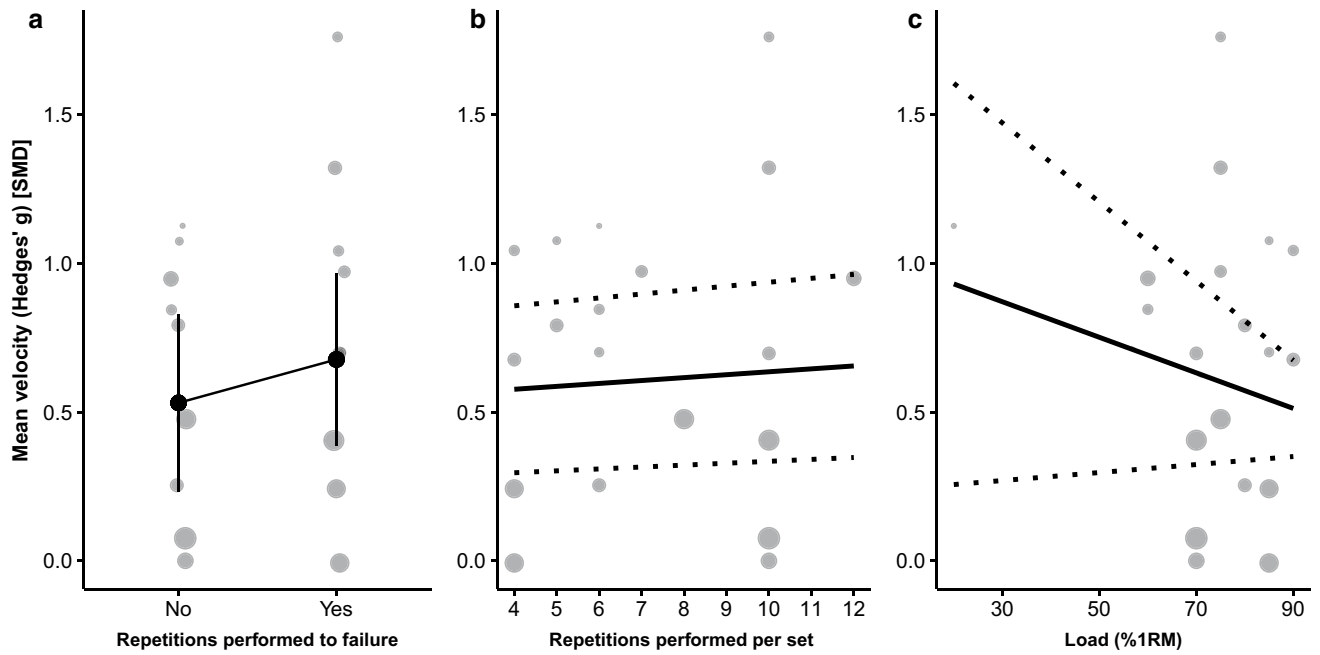


Fig. 4 Random-effects meta-regression of absolute mean velocity during alternative set structures (i.e., CS and RR combined) compared to TS while controlling for the effects of muscular failure during TS (a), number of repetitions performed during TS (b) and the load used (c). Larger data points received greater weighting than

smaller data points. Solid (non-vertical) lines represent the estimated relationship, solid vertical lines and dotted lines represent the upper and lower 95% confidence intervals. *CS* cluster set structure, *RM* repetition maximum, *RR* rest redistribution set structure, *TS* traditional set structure

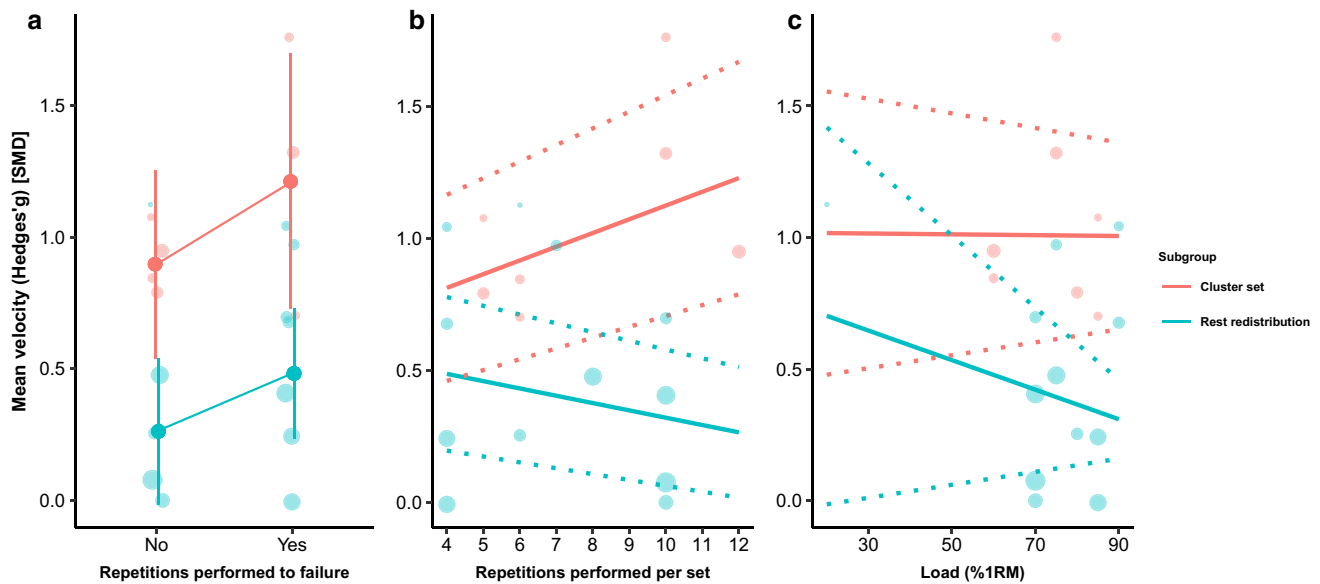
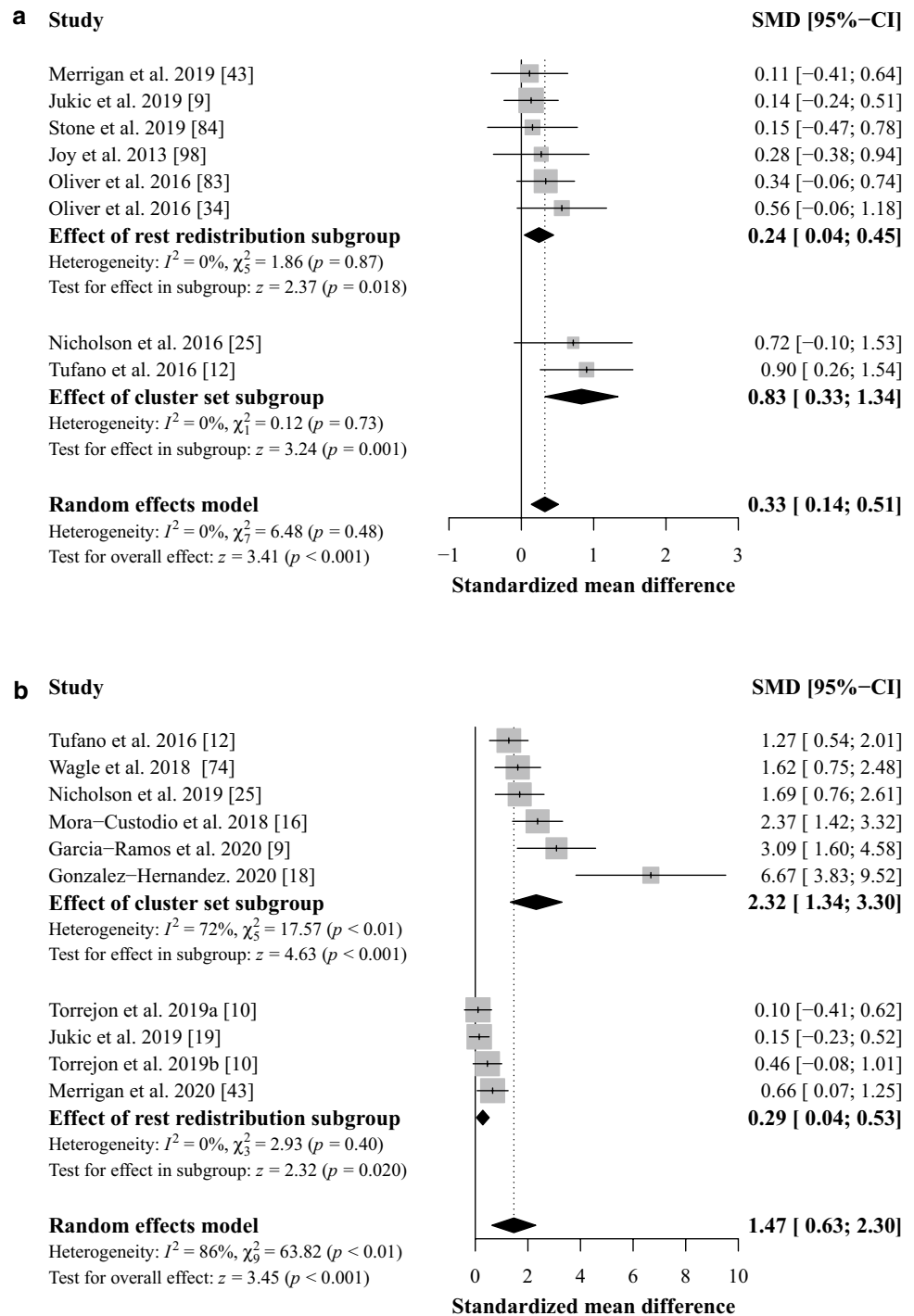


Fig. 5 Random-effects meta-regression of absolute mean velocity during alternative set structures (i.e., CS and RR separately) compared to TS while controlling for the effects of muscular failure during TS (a), number of repetitions performed during TS (b) and the load used (c). Solid red lines and circles indicates data for CS, while green lines and circles indicated data for RR. Larger data points

received greater weighting than smaller data points. Solid (non-vertical) lines represent the estimated relationship, solid vertical lines and dotted lines represent the upper and lower 95% confidence intervals. *CS* cluster set structure, *RM* repetition maximum, *RR* rest redistribution set structure, *TS* traditional set structure

Fig. 6 Random-effects meta-analysis of absolute mean power (a) and mean velocity decrement (b) during alternative set structures (i.e., CS and RR) compared to TS. Subgroup analysis based on the effects of CS and RR separately. *CI* confidence interval, *CS* cluster set structure, *RM* repetition maximum, *RR* rest redistribution set structure, *SMD* standardized mean difference, *TS* traditional set structure



resulted in significantly greater mean power maintenance when compared to TS. When the effects from alternative set structures were pooled, mean power maintenance was not significantly greater than during TS [SMD=0.78 (-0.01, 1.57); $p=0.053$; $I^2=84\%$; $k=3$]. GRADE quality of evidence for pooled results and sub-group results for RR was low (Table 2).

3.4.9 Peak Velocity Decrement

Subgroup analysis indicated that both CS [SMD=1.97 (0.14, 3.71); $p=0.02$; $I^2=81\%$; $k=2$], and RR [SMD=2.08 (0.45, 3.70); $p=0.012$; $I^2=85\%$; $k=2$] set structures resulted in significantly lower peak velocity decrement when compared to TS. When the effects from alternative set structures were pooled, peak velocity decrement was significantly lower than

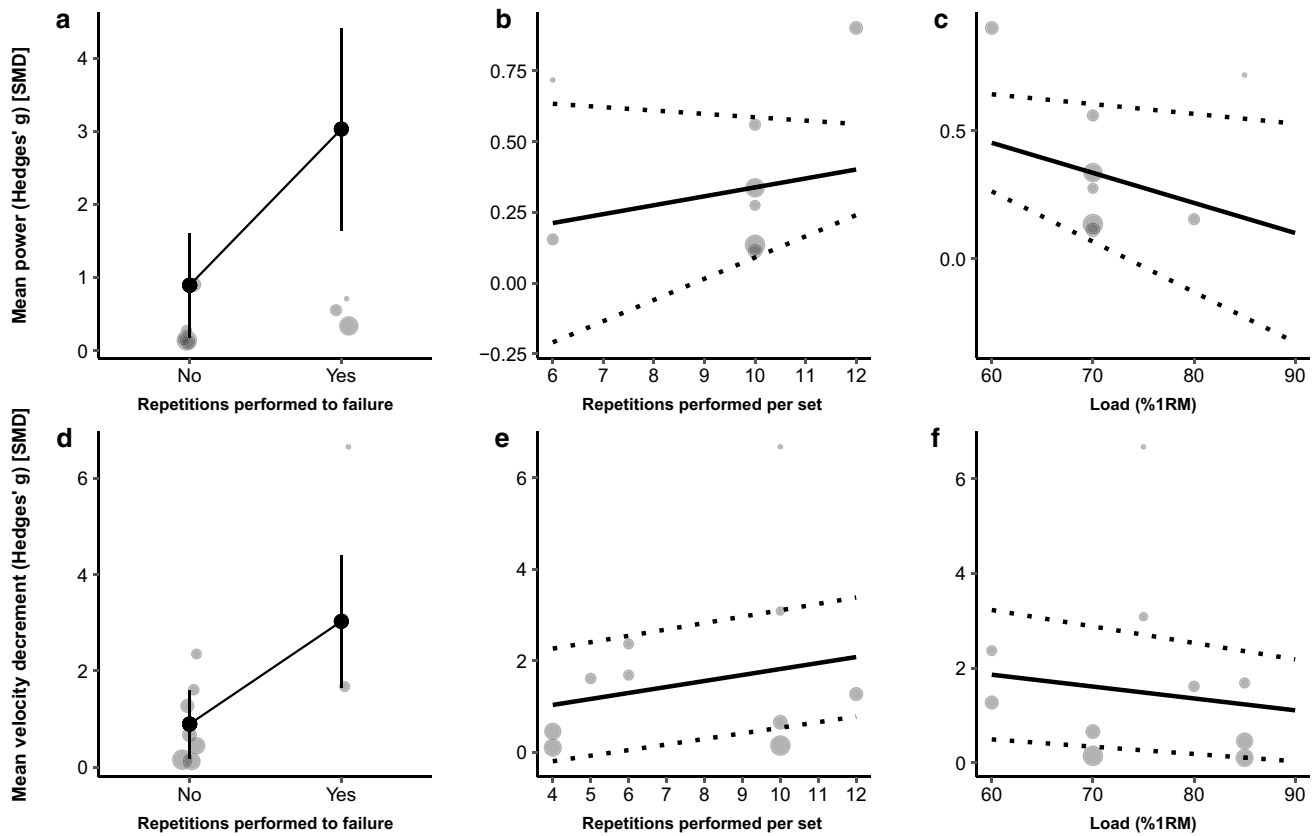


Fig. 7 Random-effects meta-regression of absolute mean power (top row) and mean velocity decrement (bottom row) during alternative set structures (i.e., CS and RR combined) compared to TS while controlling for the effects of muscular failure during TS (**a**, **d**), number of repetitions performed during TS (**b**, **e**) and the load used (**c**, **f**). Larger

data points received greater weighting than smaller data points. Solid (non-vertical) lines represent the estimated relationship, solid vertical lines and dotted lines represent the upper and lower 95% confidence intervals. CS cluster set structure, RM repetition maximum, RR rest redistribution set structure, TS traditional set structure

during TS [SMD = 1.97 (0.98, 2.95); $p < 0.001$; $I^2 = 77\%$; $k = 4$]. Pooled results and sub-group results for CS and RR all provided low GRADE quality of evidence (Table 2).

3.4.10 Peak Velocity Maintenance

Subgroup analysis indicated that both CS [SMD = 0.91 (0.39, 1.44); $p < 0.001$; $I^2 = 0\%$; $k = 2$] and RR [SMD = 2.57 (1.42, 3.73); $p < 0.001$; $I^2 = 0\%$; $k = 2$] set structures resulted in significantly greater peak velocity maintenance when compared to TS. When the effects from alternative set structures were pooled, peak velocity maintenance was significantly greater than during TS [SMD = 1.36 (0.33, 2.39); $p = 0.009$; $I^2 = 71\%$; $k = 4$]. Sub-group results for CS provided moderate whereas pooled results provided low GRADE quality of evidence (Table 2).

3.4.11 Peak Power Decrement

Subgroup analysis indicated that both CS [SMD = 1.48 (0.82, 2.13); $p < 0.001$; $I^2 = 47\%$; $k = 3$], and RR [SMD = 1.95 (0.28, 3.62); $p = 0.022$; $I^2 = 88\%$; $k = 3$] set structures resulted in significantly lower peak power decrement when compared to TS. When the effects from alternative set structures were pooled, peak power decrement was significantly lower than during TS [SMD = 1.71 (0.92, 2.49); $p < 0.001$; $I^2 = 76\%$; $k = 6$]. Sub-group results for CS provided moderate whereas pooled results and sub-group results for RR provided low GRADE quality of evidence (Table 2). The load used [$b = -0.01$ (-0.03, 0.02); $p = 0.773$] or the number of repetitions performed during TS [$b = -0.22$ (-0.48, 0.04); $p = 0.105$] were not significant moderators of the SMD for peak power decrement.

3.4.12 Peak Power Maintenance

Subgroup analysis indicated that both CS [SMD = 0.99 (0.33, 1.64); $p = 0.003$; $I^2 = 0\%$; $k = 1$], and RR [SMD = 2.09 (1.10, 3.08); $p < 0.001$; $I^2 = 0\%$; $k = 1$] set structures resulted in significantly greater peak power maintenance when compared to TS. When the effects from alternative set structures were pooled, peak power maintenance was significantly greater than during TS [SMD = 1.47 (0.40, 2.55); $p = 0.007$; $I^2 = 70\%$; $k = 2$]. GRADE quality of evidence for pooled results was low (Table 2).

3.4.13 Post-Session Jump Height Loss or Velocity Against Fixed Load Loss

Subgroup analysis indicated that CS [SMD = 0.65 (0.29, 1.01); $p < 0.001$; $I^2 = 0\%$; $k = 3$], but not RR [SMD = 0.39 (− 0.20, 0.98); $p = 0.195$; $I^2 = 0\%$; $k = 1$] set structures resulted in significantly lower post-session jump height or velocity against fixed load loss when compared to TS. When the effects from alternative set structures were pooled, post-session jump height or velocity against fixed load loss were significantly lower than during TS [SMD = 0.58 (0.27, 0.89); $p < 0.001$; $I^2 = 0\%$; $k = 4$]. GRADE quality of evidence for pooled results and sub-group results for CS was moderate (Table 2).

3.5 Metabolic Outcomes

3.5.1 Lactate

Subgroup analysis indicated that both CS [SMD = 2.17 (1.18, 3.17); $p < 0.001$; $I^2 = 67\%$; $k = 5$] and RR [SMD = 1.27 (0.87, 1.67); $p < 0.001$; $I^2 = 0\%$; $k = 3$] set structures resulted in significantly lower post-session lactate values when compared to TS (Fig. 8). When the effects from alternative set structures were pooled, lactate values were significantly lower than during TS [SMD = 1.61 (1.18, 2.04); $p < 0.001$; $I^2 = 58\%$; $k = 8$]. Sub-group results for CS provided low whereas pooled results and sub-group results for RR provided moderate GRADE quality of evidence (Table 2). Performing repetitions to failure [$b = 0.21$ (− 1.16, 1.58); $p = 0.764$] or the number of repetitions being performed during TS [$b = 0.20$ (− 0.05, 0.45); $p = 0.112$] and the load used [$b = 0.01$ (− 0.05, 0.07); $p = 0.662$] were not significant moderators of the SMD in lactate values (Fig. 9).

3.6 Perceptual Outcomes

3.6.1 Rating of Perceived Exertion

Subgroup analysis indicated that both CS [SMD = 1.14 (0.61, 1.67); $p < 0.001$; $I^2 = 0\%$; $k = 2$] and RR [SMD = 0.69

(0.27, 1.11); $p = 0.001$; $I^2 = 69\%$; $k = 4$] set structures resulted in significantly lower ratings of perceived exertion when compared to TS (Fig. 8). When the effects from alternative set structures were pooled, ratings of perceived exertion were significantly lower than during TS [SMD = 0.81 (0.45, 1.17); $p < 0.001$; $I^2 = 64\%$; $k = 6$]. Sub-group results for CS provided moderate whereas pooled results and sub-group results for RR provided low GRADE quality of evidence (Table 2). Performing repetitions to failure [$b = -0.19$ (− 0.94, 0.56); $p = 0.625$] or the number of repetitions being performed during TS [$b = 0.16$ (− 0.02, 0.33); $p = 0.08$] were not significant moderators of the SMD in rating of perceived exertion. However, the load used was a significant moderator of the SMD in rating of perceived exertion with lower loads leading to greater effects of alternative set structures on perceptual exertion [$b = -0.06$ (− 0.09, 0.02); $p = 0.004$] (Fig. 9).

4 Discussion

The present systematic review and meta-analysis is the first to synthesise the evidence regarding the efficacy of alternative set structures to alleviate mechanical, metabolic, and perceptual markers of fatigue during and after RT while also accounting for the fundamental differences between CS and RR. Based on the current evidence, the main findings of this review suggest that although both CS and RR set structures positively affect mechanical, metabolic, and perceptual responses during RT compared to TS, the magnitude of these positive effects is generally much greater for CS than for RR. Therefore, the alteration of TS to a different alternative set structure (CS or RR) is expected to provide different acute training stimuli, which further highlights the need for a clearer distinction between the structures.

4.1 Mechanical Outcomes

4.1.1 Mean and Peak Velocity and Power (Overall for Entire Protocol)

Moderate to high GRADE quality evidence indicated significant differences between alternative set structures and TS in absolute mean and peak velocity and power. When comparing TS against CS and RR separately, similar moderate to high GRADE quality of evidence was provided. However, the positive effects of CS over TS were much larger (SMD: 0.54–1.01) than those observed for RR (SMD: − 0.01 to 0.39). Therefore, the extra intra-set rest periods during CS likely allow for immediate energy stores and subsequent performance to be better maintained compared to RR. Furthermore, although RR positively affected both peak power (SMD: 0.30) and peak velocity (SMD:

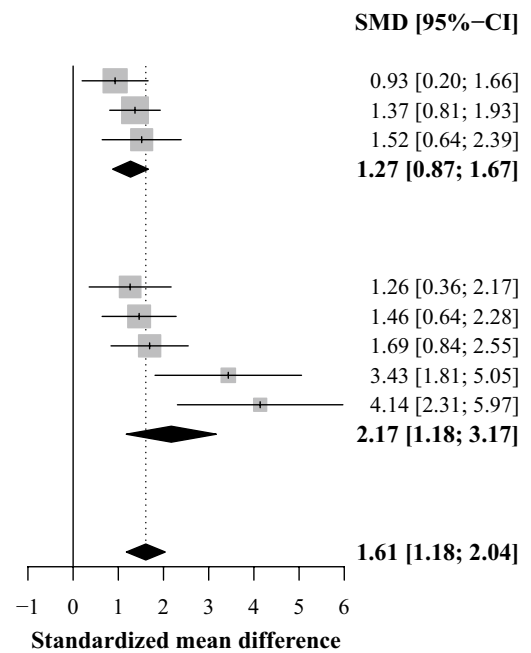
Fig. 8 Random-effects meta-analysis of blood lactate (a) and rating of perceived exertion (b) during alternative set structures (i.e., CS and RR) compared to TS. Subgroup analysis based on the effects of CS and RR separately. *CI* confidence interval, *SMD* standardized mean difference. *CS* cluster set structure, *RR* rest redistribution set structure, *TS* traditional set structure

a Study

Denton et al. 2006 [97]
Oliver et al. 2015 [70]
Iglesias-Soler et al. 2012 [17]
Effect of rest redistribution subgroup
Heterogeneity: $I^2 = 0\%$, $\chi^2_2 = 1.27$ ($p = 0.53$)
Test for effect in subgroup: $z = 6.29$ ($p < 0.001$)

Nicholson et al. 2016 [25]
Girman et al. 2014 [75]
Mora-Custodio et al. 2018 [16]
Garcia-Ramos et al. 2020 [9]
Gonzalez-Hernandez. 2020 [18]
Effect of cluster set subgroup
Heterogeneity: $I^2 = 67\%$, $\chi^2_4 = 12.25$ ($p = 0.02$)
Test for effect in subgroup: $z = 4.27$ ($p < 0.001$)

Random effects model
Heterogeneity: $I^2 = 58\%$, $\chi^2_7 = 16.49$ ($p = 0.02$)
Test for overall effect: $z = 7.32$ ($p < 0.001$)

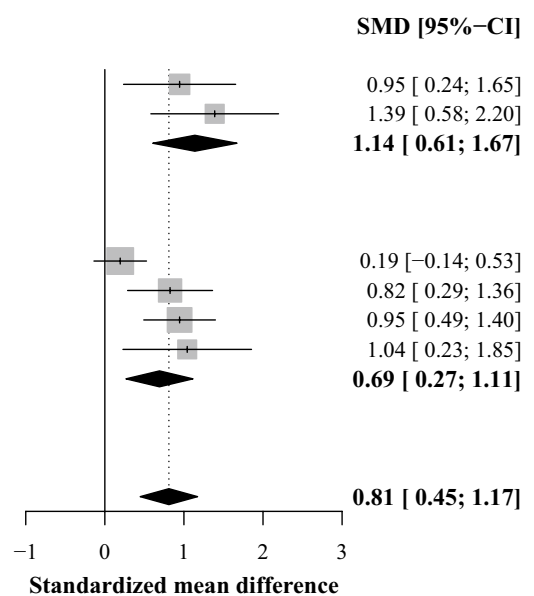


b Study

Hardee et al. 2012 [20]
Gonzalez-Hernandez. 2020 [18]
Effect of cluster set subgroup
Heterogeneity: $I^2 = 0\%$, $\chi^2_1 = 0.66$ ($p = 0.42$)
Test for effect in subgroup: $z = 4.19$ ($p < 0.001$)

Hoo et al. 2019 [72]
Mayo et al. 2019 [21]
Jukic et al. 2019 [19]
Mayo et al. 2014 [22]
Effect of rest redistribution subgroup
Heterogeneity: $I^2 = 69\%$, $\chi^2_3 = 9.71$ ($p = 0.02$)
Test for effect in subgroup: $z = 3.19$ ($p = 0.001$)

Random effects model
Heterogeneity: $I^2 = 64\%$, $\chi^2_3 = 14.04$ ($p = 0.02$)
Test for overall effect: $z = 4.37$ ($p < 0.001$)



0.28), this effect was not significant ($p < 0.05$), indicating that CS should be preferred when the aim is to have a greater overall peak velocity and peak power during RT (Fig. 3). While it may be tempting to conclude that RR is not effective in enhancing peak velocity and power, it must be noted that considerably fewer studies report peak variables as opposed to mean variables, which may have affected the results (i.e., lower statistical power and hence stability of the results). This is further supported by the fact that beneficial effects of CS and RR were greater for the mean variables, again, likely due to the higher number of studies reporting mean velocity and power.

Furthermore, neither the presence of muscular failure during TS, the number of total repetitions being performed, nor the load used were significant moderators of the differences between TS and alternative set structures (both combined and separately) (Figs. 4, 5). However, there was still a tendency to show greater effects of RR and CS for studies in which muscular failure was reached during TS. This probably explains the findings from more recent studies suggesting that when TS are not performed to, or near to, muscular failure, the effects of alternative set structures on absolute measures of mechanical performance are not as profound [19, 33, 77, 78]

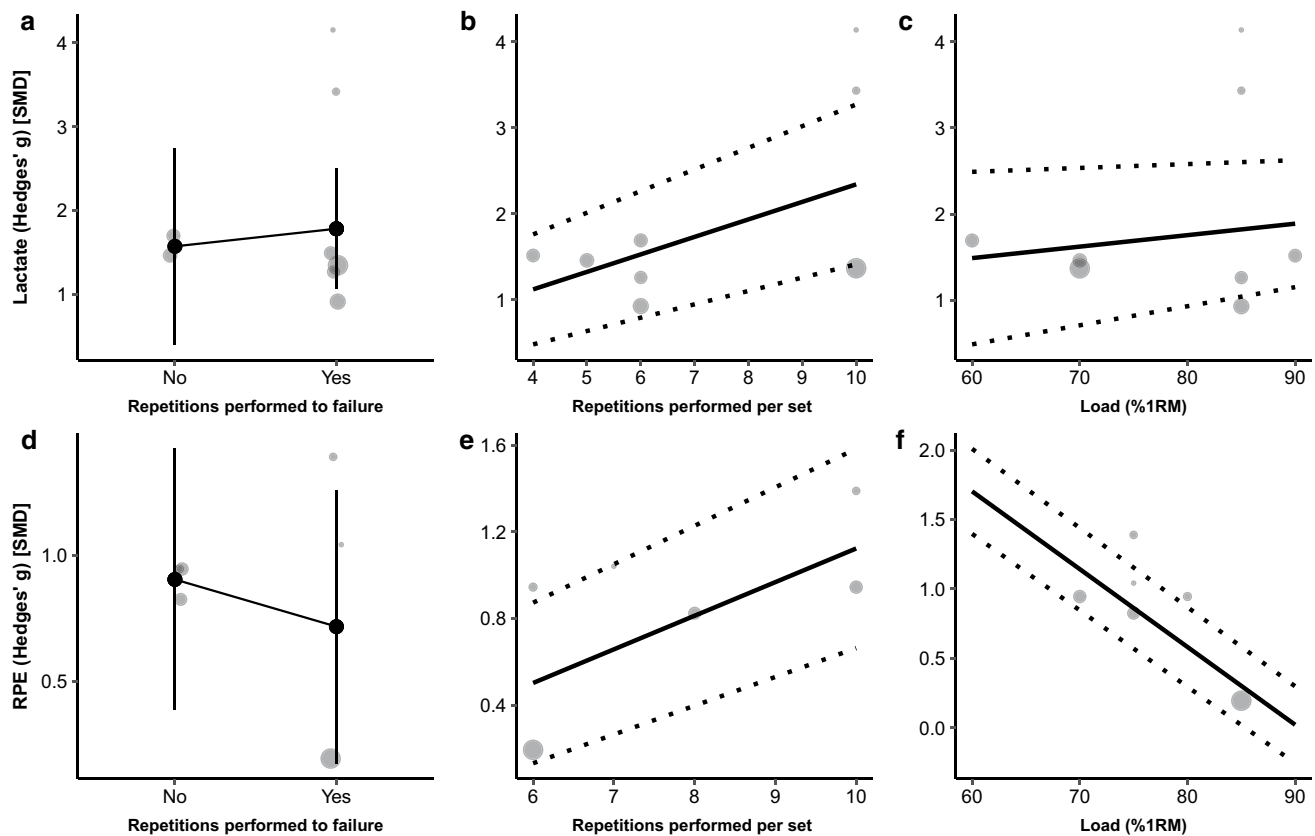


Fig. 9 Random-effects meta-regression of blood lactate (top row) and rating of perceived exertion (bottom row) during alternative set structures (i.e., CS and RR combined) compared to TS while controlling for the effects of muscular failure during TS (**a**, **d**), number of repetitions performed during TS (**b**, **e**) and the load used (**c**, **f**). Larger

data points received greater weighting than smaller data points. Solid (non-vertical) lines represent the estimated relationship, solid vertical lines and dotted lines represent the upper and lower 95% confidence intervals. CS cluster set structure, RM repetition maximum, RR rest redistribution set structure, TS traditional set structure

4.1.2 Mean and Peak Velocity and Power Decrement (Loss Throughout the Protocol)

Although analysing mean and peak velocity and power provides an overview of what happens during TS, CS, and RR, such an overarching approach does not explain what actually happens during exercise, as one continues to perform more repetitions. Therefore, many studies also calculate the decrement of the mechanical variables from the beginning to the end of a set or protocol, which could arguably be seen as a true indicator of fatigue since it represents a change in velocity or power as fatigue manifests. Our meta-analyses provided low to moderate GRADE evidence, indicating positive effects of alternative set structures on mean and peak velocity and power decrement (i.e. the decrease in performance throughout the protocol was not as much as in TS). Specifically, the effects of CS on mean velocity decrement were much greater (SMD: 2.32) compared to RR (SMD: 0.29). However, for mean power and peak velocity decrement, the positive effects of CS and RR over TS were comparable, with RR allowing for even less peak power decrement when

compared to TS and CS. The discrepancy between the findings could likely be explained by the varying number of studies reporting decrement variables and the high heterogeneity that affected the results of the subgroup analyses.

Regardless of the decrement variable being calculated, the beneficial effects of alternative set structures (both combined and separately) were always greater than, or varied less than, absolute mean and peak velocity and power variables (i.e., overall values for entire the protocol). Thus, the greater and more consistent effect of RR and CS on velocity and power loss suggests that “absolute” and “decrement” variables should be classified independently. This is further supported by numerous studies, which found how the effects of CS and RR were much smaller on absolute rather than decrement mechanical outcomes [12, 14, 43, 44]. Furthermore, performing repetitions to failure during TS was a significant moderator of the differences between alternative set structures and TS (i.e. less mean velocity loss in CS and RR compared to TS to failure). Although interesting, this finding likely represents a type I error for the following reasons: TS were performed to failure in only 3 out of 10 studies in

the meta-analysis, all of which compared TS to CS, and we found no significant association between the effects of alternative set structures and whether repetitions were performed to muscular failure when analysing CS separately. Therefore, we suspect this analysis to be more prone to inferential errors and thus, to represent a false positive finding given the non-significant effects of muscular failure with a more equal distribution (3 vs 3) of “failure” and “non-failure” studies that compared TS and CS only.

4.1.3 Mean and Peak Velocity and Power Maintenance (Overall for Entire Protocol)

Meta-analyses in the present review provided low to moderate GRADE evidence that alternative set structures allow for greater maintenance (SMD: 0.95–1.47) of mean velocity, peak velocity, and peak power, but not mean power, during RT. Although few studies reported these mechanical outcomes in the present review, mean power maintenance was also affected by conflicting findings of the only two studies that reported on this outcome. More specifically, while Merrigan et al. [43] found RR to be effective at maintaining mean power during RT, Jukic et al. [19] reported no significant differences between TS and RR. The discrepancy between findings of these two studies could likely be explained by the differences in the inter-set rest time allowed for the TS group (2 vs 4 min). Similar to other mechanical outcomes, CS were more effective than RR at maintaining mechanical performance during RT.

While it may not seem obvious how calculating the decline and the maintenance of mechanical variables could provide a different insight into a training stimulus imposed by altering the set structure during RT, several studies clearly indicated this to be the case [12, 19, 43]. For instance, Tufano et al. [12] showed that TS resulted in a 23% decline in velocity and power from the first repetition to the last, but when the maintenance variable was calculated, which takes into account all repetitions not just the first and the last, TS resulted in a “decrease” of only 8%. Therefore, it is important to note that absolute, decline and maintenance variables of mechanical outcomes may each tell a different story and should, therefore, be classified independently. The emphasis on each of them (combined or in isolation) should probably depend upon the objective of RT. For instance, if power adaptations are desired, decrement variables could be used to quantify a decline in performance from the first repetition to the last, and to detect when the decline started to affect performance (i.e., when calculated for each repetition). On the other hand, maintenance variables could provide a greater insight into the overall degree of fatigue induced by RT when considering all the repetitions in a given set. Evidently, both decline and maintenance variables provide

distinct information which can then better inform future training decisions.

4.1.4 Performance Before and After the Entire Protocol

While the results of the present review suggest that alternative set structures are effective at maintaining acute RT performance, it is also important to consider the residual mechanical fatigue induced by RT sessions. For instance, there are many sports (e.g., team-sports) in which RT precedes sport-specific training. In that regard, alternative set structures might also enable less interference with subsequent sport training. Moderate GRADE evidence showed lower mechanical fatigue levels after RT in which alternative set structures were used, as indicated by less jump height loss or less velocity loss against a fixed load (SMD: 0.58). It is important to note that the magnitude of these effects was much lower when compared to the effects on mechanical fatigue assessed during actual training (i.e., decline and maintenance variables) suggesting that the rate of post-session recovery might not be as affected by alteration of the set structure. This finding is in agreement with a recent study published after the present review was conducted [44], which showed that RR was more effective in maintaining force production during an isokinetic unilateral knee extension at different velocities than inducing faster peripheral recovery after training. However, since there were a limited number of studies in the present review, which varied in exercise selection, load, and number of sets and repetitions, future research is needed to determine if alternative set structures impact mechanical markers of fatigue differentially during and after RT.

4.2 Metabolic and Perceptual Outcomes

Moderate GRADE evidence indicated significantly lower blood lactate concentrations during alternative set structures when compared to TS (SMD: 1.61). Similar to all mechanical outcomes, when the effects of CS and RR were analysed separately, CSs were more effective than RR in reducing the metabolic stress imposed by RT since much lower blood lactate concentrations were generally observed after CS (SMD: 2.12) than RR (SMD: 1.12) when compared to TS (Fig. 8). This contention is supported by the literature since various mechanical markers of fatigue have been associated with lactate accumulation during RT [15, 79–81]. Based on earlier research by Gorostiaga et al. [38–40] and Bogdanis et al. [35–37], it is logical that CS inclusive of extra rest periods were effective at reducing metabolic stress during RT. However, it is interesting that RR was also highly effective for managing metabolic fatigue during RT, despite the total rest time not increasing compared to TS. Furthermore, meta-regressions suggest

that alternative set structures are effective in reducing metabolic stress compared to TS, irrespective of the load used, the number of repetitions performed, or whether repetitions were performed to failure during TS (Fig. 9). Collectively, this suggests that higher metabolic stress induced by TS might negatively affect performance of subsequent exercises in a session, which could be alleviated using alternative set structures.

Low to moderate GRADE evidence indicates that both CS and RR were effective at reducing perceived exertion after RT compared to TS (Fig. 8). As with every other outcome in this review, CS were more effective than RR, which is not surprising considering the strong relationship between the mechanical, metabolic, and perceptual markers of fatigue reviewed and meta-analysed in this paper [9, 15, 16, 20, 23, 82]. Meta-regressions showed that performing repetitions to muscular failure during TS and the number of repetitions performed were not significant moderators of the magnitude of the effect on RPE (Fig. 9). Interestingly, however, meta-regression also showed that lower loads were associated with stronger effects of alternative set structures on RPE. Although this might sound counterintuitive on the surface, the range of loads included in the analysis was narrow (i.e., 70–85%), and studies using the highest loads showed the smallest effect. In addition, due to its sample size, one study [72] in which repetitions were performed to failure regardless of the set structure carried greater weight in this analysis. Additionally, only 6 repetitions were performed during TS in this study whereas more than 8 repetitions were performed during TS in other studies included in the analysis. Consequently, this study [72] also affected the relationship between the number of repetitions performed and the effect of alternative set structures on RPE, which was non significant. Since this single study [72] was at a high risk of bias, the number of repetitions performed during TS likely was a significant moderator of the effects of alternative set structures on RPE, with more repetitions per set leading to greater differences between the set structures, whereas the load was not.

4.3 Implications for Training and Research

Overall, the findings of this review suggest that alternative set structures are effective at ameliorating acute mechanical fatigue during and after RT as well as metabolic, and perceptual fatigue after RT. However, greater differences were observed between CS and TS than RR and TS for most outcomes, confirming that CS and RR likely provide different training stimuli and subsequently may affect recovery in different ways. This is not surprising, considering that when compared to TS, CS structures contain extra rest while RR structures do not, further highlighting that CS and RR should be classified independently. Although on the surface

it may seem obvious to classify CS and RR as two different types of alternative set structures (due to the extra total rest in CS), the term “cluster set” has been used loosely in the literature, resulting in an umbrella term that encompasses different types of set structures [41, 44, 45]. As such, the results of investigations are not always in agreement, which might leave sport professionals unsure of how they should construct set structures in practice. Therefore, we urge future investigators to consider the fundamental differences between alternative set structures. Doing so will allow researchers and sport professionals to compare and contrast different set structure designs with more accuracy and less confusion.

Conclusions regarding the effects of alternative set structures are often based on absolute mechanical performance (e.g., mean and peak velocity and power averaged across the whole session) [45] or the decline and maintenance of these mechanical outcomes during RT [12, 33, 43, 44]. While studies often reach similar conclusions to varying degrees of magnitude, it must be noted that analysing the absolute mean and peak values of a given mechanical variable is not the same as assessing the change in performance throughout the RT session. The findings of this review support this contention, showing greater effects of alternative set structures on decline and maintenance, than on absolute mechanical values during RT. While this might partially be explained by the design of a given set structure, the velocity or power of the initial repetitions of any set structure will most likely be similar, attenuating differences between TS and alternative set structures in the overall response [15, 70, 83, 84]. However, although the differences between TS and CS or RR become more evident when considering decline or maintenance calculations, the differences become even more apparent as the number of repetitions performed during a set increase. For instance, CS can result in greater absolute velocity and power compared to TS [12], but much larger differences can be observed if velocity and power maintenance (i.e., considering all the repetitions within sets) are calculated. In addition, a decline in velocity and power (i.e., considering only the last repetition relative to the first) is even more sensitive in detecting fatigue-induced changes in mechanical performance with the greatest differences typically being observed between TS and CS in decline variables. This suggests, as individual studies have hypothesised, that alternative set structures might not be as effective for producing higher absolute velocity or power outputs, but rather, better serve to reduce fatigue [43]. Therefore, it is evident that the absolute, maintenance, and decline of mechanical outcomes may each tell a different story and hence, should be classified independently. By utilising all three during RT, sport professionals will gain a greater insight into: (1) overall mechanical performance; (2) degree of neuromuscular fatigue induced by a given set; (3) when

the fatigue-induced changes become apparent, and at what rate they develop.

Similar to mechanical fatigue during RT, alternative set structures were also effective at alleviating post-RT mechanical fatigue. While these effects were generally smaller and suggest that alternative set structures might not be as effective in combating acute residual fatigue induced by RT, research on this topic is lacking. Therefore, future research is needed to explore the effectiveness of alternative set structures in mitigating residual fatigue after RT. On the other hand, results from this meta-analysis suggest that alternative set structures are highly effective at reducing metabolic stress and perceptual exertion induced by RT when compared to TS. Since low RPE and blood lactate values have been associated with higher velocity and power outputs [9, 18, 20, 85], alternative set structures might be effective not only in stimulating strength-speed and speed-strength adaptations, but also in reducing recovery time following RT [86–90]. This would be valuable for sport professionals, especially those working in team-sport settings where competitions are held every weekend or even every 3–4 days who need to manage RT fatigue for both game performance and sport specific training sessions [91–93]. In addition, since sport professionals working in those settings are often time-constrained, RR might be preferable to CS since they do not extend total training time while still aiding fatigue management during and after RT. Furthermore, the metabolic and perceptual outcomes, coupled with mechanical performance findings, could also imply that alternative set structures may enable greater volumes of work to be completed by allowing more sets and repetitions to be performed [33, 73, 83] or greater loads [13, 25] to be lifted which can ultimately result in greater strength adaptations as well as an increase in lean body mass [7, 26, 94–96]. For these reasons, CS or RR set structures provide sport professionals with a multitude of options to design their training to better manage fatigue while inducing the desired adaptations.

4.4 Limitations and Considerations

Several aspects of this review should be considered when interpreting the findings. First, the small number of studies including female participants limits the generalizability to female participants, and more research should include females when possible. Second, most meta-analyses of decline and maintenance of mechanical outcomes in the present review were affected by the high levels of heterogeneity. While we attempted to explore the cause of the heterogeneity by performing sub-group analyses and meta-regressions when sufficient studies were available, it is likely the small number of studies was the primary contributor to the observed heterogeneity. In addition, general disagreement

between the magnitudes of the effect of CS and RR on some of the outcomes consequently affected the heterogeneity of the pooled results, which can rather be seen as a strength than a limitation since the sub-group analysis was effective at identifying the source of heterogeneity. Third, while we performed meta-regression to examine how muscular failure, number of repetitions, and load used during TS impacted the effectiveness of alternative set structures, compared to TS, the number of included studies only allowed this to be performed on several pooled results and only a few sub-group results. In addition, for some of the outcomes, the range of loads was narrow, and studies in which TS were performed to failure were not equally distributed among the subgroups, limiting the generalizability of the meta-regression's findings. Therefore, future studies should compare TS and alternative set structures across a wide range of loads. This is important since one might argue that more frequent racking and un-racking the barbell with very heavy loads (e.g., > 85% 1RM) might negatively affect RT performance. It is also important to note that other studies contain valuable data that did not fit within the scope of the present review, which should still be considered when designing RT programs. Finally, due to insufficient information from individual studies in this review, a default correlation coefficient of 0.50 was used for all studies, which could have caused an underestimation of the actual differences between TS and alternative set structures.

5 Conclusion

The results of the present systematic review and meta-analysis suggest that alternative set structures are very effective at reducing mechanical fatigue, perceptual exertion, and metabolic stress during and after RT. However, fundamental differences in total rest time result in CS being generally more effective than RR for alleviating fatigue during RT. These fundamental differences highlight the importance of classifying alternative set structures independently. In addition, absolute, decline, and maintenance of mechanical variables during RT, as well as residual mechanical fatigue after RT, are all affected differently by alternative set structures, suggesting that each variable may have different implications. Therefore, these variables should be assessed independently and not be used interchangeably when evaluating the effectiveness of alternative set structures. With all of this in mind, researchers, sport professionals, and athletes should take these factors into consideration when choosing to implement CS, RR, or any form of alternative set structure in research and practice.

Compliance with Ethical Standards

Author contributions JJ performed the meta-analysis and meta-regression and wrote the first draft of the manuscript. All authors edited and revised the manuscript and approved the final version of the manuscript.

Funding No external sources of funding were used to assist in the preparation of this article.

Conflict of interest Ivan Jukic, Amador García Ramos, Eric Helms, Michael McGuigan and James Tufano declare that they have no conflicts of interest relevant to the content of this review.

Availability of data and material The datasets generated during and/or analysed during the current review are available from the corresponding author on reasonable request.

Ethics approval Not applicable.

Consent Not applicable.

References

- Suchomel TJ, Nimphius S, Bellon CR, Stone MH. The importance of muscular strength: training considerations. *Sports Med.* 2018;48(4):765–85.
- Suchomel TJ, Nimphius S, Stone MH. The importance of muscular strength in athletic performance. *Sports Med.* 2016;46(10):1419–49.
- Kraemer WJ, Ratamess NA, French DN. Resistance training for health and performance. *Curr Sports Med Rep.* 2002;1(3):165–71.
- O'Connor PJ, Herring MP, Caravallho A. Mental health benefits of strength training in adults. *Am J Lifestyle Med.* 2010;4(5):377–96.
- Feigenbaum MS, Pollock ML. Prescription of resistance training for health and disease. *Med Sci Sports Exerc.* 1999;31(1):38–45.
- Grgic J, Lazinica B, Mikulic P, Krieger JW, Schoenfeld BJ. The effects of short versus long inter-set rest intervals in resistance training on measures of muscle hypertrophy: a systematic review. *Eur J Sport Sci.* 2017;17(8):983–93.
- Grgic J, Schoenfeld BJ, Davies TB, Lazinica B, Krieger JW, Pedisic Z. Effect of resistance training frequency on gains in muscular strength: a systematic review and meta-analysis. *Sports Med.* 2018;48(5):1207–20.
- Grgic J, Schoenfeld BJ, Skrepnik M, Davies TB, Mikulic P. Effects of rest interval duration in resistance training on measures of muscular strength: a systematic review. *Sports Med.* 2018;48(1):137–51.
- García-Ramos A, González-Hernández JM, Baños-Pelegrín E, Castaño-Zambudio A, Capelo-Ramírez F, Boullosa D, et al. Mechanical and metabolic responses to traditional and cluster set configurations in the bench press exercise. *J Strength Cond Res.* 2020;34(3):663–70.
- Torrejón A, Janicijevic D, Haff GG, García-Ramos A. Acute effects of different set configurations during a strength-oriented resistance training session on barbell velocity and the force-velocity relationship in resistance-trained males and females. *Eur J Appl Physiol.* 2019;119(6):1409–17.
- Hardee JP, Triplett NT, Utter AC, Zwetsloot KA, McBride JM. Effect of interrepetition rest on power output in the power clean. *J Strength Cond Res.* 2012;26(4):883–9.
- Tufano JJ, Conlon JA, Nimphius S, Brown LE, Seitz LB, Williamson BD, et al. Maintenance of velocity and power with cluster sets during high-volume back squats. *Int J Sports Physiol Perform.* 2016;11(7):885–92.
- Tufano JJ, Conlon JA, Nimphius S, Brown LE, Banyard HG, Williamson BD, et al. Cluster sets: permitting greater mechanical stress without decreasing relative velocity. *Int J Sports Physiol Perform.* 2017;12(4):463–9.
- Tufano JJ, Conlon JA, Nimphius S, Brown LE, Petkovic A, Frick J, et al. Effects of cluster sets and rest-redistribution on mechanical responses to back squats in trained men. *J Hum Kinet.* 2017;58(1):35–43.
- Tufano JJ, Conlon JA, Nimphius S, Oliver JM, Kreutzer A, Haff GG. Different cluster sets result in similar metabolic, endocrine, and perceptual responses in trained men. *J Strength Cond Res.* 2019;33(2):346–54.
- Mora-Custodio R, Rodríguez-Rosell D, Yáñez-García JM, Sánchez-Moreno M, Pareja-Blanco F, González-Badillo JJ. Effect of different inter-repetition rest intervals across four load intensities on velocity loss and blood lactate concentration during full squat exercise. *J Sports Sci.* 2018;36(24):2856–64.
- Iglesias-Soler E, Carballeira E, Sánchez-Otero T, Mayo X, Jiménez A, Chapman ML. Acute effects of distribution of rest between repetitions. *Int J Sports Med.* 2012;33(5):351–8.
- González-Hernández JM, García-Ramos A, Castaño-Zambudio A, Capelo-Ramírez F, Marquez G, Boullosa D, et al. Mechanical, metabolic, and perceptual acute responses to different set configurations in full squat. *J Strength Cond Res.* 2020;34(6):1581–90.
- Jukic I, Tufano JJ. Shorter but more frequent rest periods: no effect on velocity and power compared to traditional sets not performed to failure. *J Hum Kinet.* 2019;66:257–68.
- Hardee JP, Lawrence MM, Utter AC, Triplett NT, Zwetsloot KA, McBride JM. Effect of inter-repetition rest on ratings of perceived exertion during multiple sets of the power clean. *Eur J Appl Physiol.* 2012;112(8):3141–7.
- Mayo X, Iglesias-Soler E, Kingsley JD. Perceived exertion is affected by the submaximal set configuration used in resistance exercise. *J Strength Cond Res.* 2019;33(2):426–32.
- Mayo X, Iglesias-Soler E, Fernández-Del-Olmo M. Effects of set configuration of resistance exercise on perceived exertion. *Percept Mot Skills.* 2014;119(3):825–37.
- Jukic I, Tufano JJ. Acute effects of shorter but more frequent rest periods on mechanical and perceptual fatigue during a weightlifting derivative at different loads in strength-trained men. *Sports Biomech.* 2020. <https://doi.org/10.1080/14763141.2020.1747530>.
- Morales-Artacho AJ, Padial P, García-Ramos A, Pérez-Castilla A, Feriche B. Influence of a cluster set configuration on the adaptations to short-term power training. *J Strength Cond Res.* 2018. <https://doi.org/10.1519/jsc.0000000000001925>.
- Nicholson G, Ispoglou T, Bissas A. The impact of repetition mechanics on the adaptations resulting from strength-, hypertrophy- and cluster-type resistance training. *Eur J Appl Physiol.* 2016;116(10):1875–88.
- Oliver JM, Jagim AR, Sanchez AC, Mardock MA, Kelly KA, Meredith HJ, et al. Greater gains in strength and power with intra-set rest intervals in hypertrophic training. *J Strength Cond Res.* 2013;27(11):3116–311.
- Haff G, Nimphius S. Training principles for power. *Strength Cond J.* 2012;34(6):2–12.
- Suchomel TJ, Comfort P, Lake JP. Enhancing the force-velocity profile of athletes using weightlifting derivatives. *Strength Cond J.* 2017;39(1):10–20.
- Suchomel TJ, Comfort P, Stone MH. Weightlifting pulling derivatives: rationale for implementation and application. *Sports Med.* 2015;45(6):823–39.

30. Pareja-Blanco F, Rodríguez-Rosell D, Sánchez-Medina L, Gorostiaga E, González-Badillo J. Effect of movement velocity during resistance training on neuromuscular performance. *Int J Sports Med.* 2014;35(11):916–24.
31. Padulo J, Mignogna P, Mignardi S, Tonni F, D'ottavio S. Effect of different pushing speeds on bench press. *Int J Sports Med.* 2012;33(05):376–80.
32. González-Badillo JJ, Rodríguez-Rosell D, Sánchez-Medina L, Gorostiaga EM, Pareja-Blanco F. Maximal intended velocity training induces greater gains in bench press performance than deliberately slower half-velocity training. *Eur J Sports Sci.* 2014;14(8):772–81.
33. Tufano JJ, Halaj M, Kampmiller T, Novosad A, Buzgo G. Cluster sets vs traditional sets: levelling out the playing field using a power-based threshold. *PLoS ONE.* 2018;13(11):e0208035.
34. Oliver JM, Jenke SC, Mata JD, Kreutzer A, Jones MT. Acute effect of cluster and traditional set configurations on myokines associated with hypertrophy. *Int J Sports Med.* 2016;37(13):1019–24.
35. Bogdanis GC, Nevill ME, Boobis LH, Lakomy H. Contribution of phosphocreatine and aerobic metabolism to energy supply during repeated sprint exercise. *J Appl Physiol.* 1996;80(3):876–84.
36. Bogdanis G, Nevill M, Lakomy H, Boobis L. Power output and muscle metabolism during and following recovery from 10 and 20 s of maximal sprint exercise in humans. *Acta Physiol Scand.* 1998;163(3):261–72.
37. Bogdanis GC, Nevill ME, Boobis LH, Lakomy H, Nevill AM. Recovery of power output and muscle metabolites following 30 s of maximal sprint cycling in man. *J Physiol.* 1995;482(2):467–80.
38. Gorostiaga EM, Navarro-Amézqueta I, Calbet JA, Hellsten Y, Cusso R, Guerrero M, et al. Energy metabolism during repeated sets of leg press exercise leading to failure or not. *PLoS ONE.* 2012;7(7):e40621.
39. Gorostiaga EM, Navarro-Amézqueta I, Calbet JA, Sánchez-Medina L, Cusso R, Guerrero M, et al. Blood ammonia and lactate as markers of muscle metabolites during leg press exercise. *J Strength Cond Res.* 2014;28(10):2775–85.
40. Gorostiaga EM, Navarro-Amézqueta I, Cusso R, Hellsten Y, Calbet JA, Guerrero M, et al. Anaerobic energy expenditure and mechanical efficiency during exhaustive leg press exercise. *PLoS ONE.* 2010;5(10):e13486.
41. Tufano JJ, Brown LE, Haff GG. Theoretical and practical aspects of different cluster set structures: a systematic review. *J Strength Cond Res.* 2017;31(3):848–67.
42. Jukic I, Tufano JJ. Rest redistribution functions as a free and ad-hoc equivalent to commonly used velocity-based training thresholds during clean pulls at different loads. *J Hum Kinet.* 2019;68:5.
43. Merrigan JJ, Tufano JJ, Oliver JM, White JB, Fields JB, Jones MT. Reducing the loss of velocity and power in women athletes via rest redistribution. *Int J Sports Physiol Perform.* 2020;15(2):255–61.
44. Tufano JJ, Omckirk D, Malecek J, Pisz A, Halaj M, Scott BR. Traditional sets vs rest-redistribution: a laboratory-controlled study of a specific cluster set configuration at fast and slow velocities. *Appl Physiol Nutr Med.* 2019;45(4):421–330.
45. Latella C, Teo W-P, Drinkwater EJ, Kendall K, Haff GG. The acute neuromuscular responses to cluster set resistance training: a systematic review and meta-analysis. *Sports Med.* 2019;49(12):1861–77.
46. Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Syst Rev.* 2015;4(1):1.
47. Higgins JP, Altman DG, Gøtzsche PC, Jüni P, Moher D, Oxman AD, et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *BMJ.* 2011;343:d5928.
48. Miller JR, Van Hooren B, Bishop C, Buckley JD, Willy RW, Fuller JT. A systematic review and meta-analysis of crossover studies comparing physiological, perceptual and performance measures between treadmill and overground running. *Sports Med.* 2019;49(5):763–82.
49. Van Hooren B, Fuller JT, Buckley JD, Miller JR, Sewell K, Rao G, et al. Is motorized treadmill running biomechanically comparable to overground running? A systematic review and meta-analysis of cross-over studies. *Sports Med.* 2019;50(4):785–813.
50. Davies TB, Kuang K, Orr R, Halaki M, Hackett D. Effect of movement velocity during resistance training on dynamic muscular strength: a systematic review and meta-analysis. *Sports Med.* 2017;47(8):1603–17.
51. Atkins D, Best D, Briss P, Eccles M, Falck-Ytter Y, Flottorp S, et al. Grading quality of evidence and strength of recommendations. *BMJ.* 2004;328(7454):1490.
52. Viechtbauer W. Conducting meta-analyses in R with the metafor package. *J Stat Softw.* 2010;36(3):1–48.
53. Schwarzer G, Carpenter JR, Rücker G. Meta-analysis with R. Berlin: Springer; 2015.
54. R Core Team. R: a language and environment for statistical computing. R Foundation for Statistical Computing, Vienna, Austria; (2018). <https://www.R-project.org/>. Accessed 26 May 2020.
55. Veroniki AA, Jackson D, Viechtbauer W, Bender R, Bowden J, Knapp G, et al. Methods to estimate the between-study variance and its uncertainty in meta-analysis. *Res Synth Methods.* 2016;7(1):55–79.
56. Borenstein M, Hedges LV, Higgins JP, Rothstein HR. Introduction to meta-analysis. New York: Wiley; 2011.
57. Elbourne DR, Altman DG, Higgins JP, Curtin F, Worthington HV, Vail A. Meta-analyses involving cross-over trials: methodological issues. *Int J Epidemiol.* 2002;31(1):140–9. <https://doi.org/10.1093/ije/31.1.140>.
58. Cohen J. The concepts of power analysis. Statistical power analysis for the behavioral sciences. Hillsdale: L. Erlbaum Associates; 1988. p. 1–17.
59. Fu R, Gartlehner G, Grant M, Shamliyan T, Sedrakyan A, Wilt TJ, et al. Conducting quantitative synthesis when comparing medical interventions: AHRQ and the Effective Health Care Program. *J Clin Epidemiol.* 2011;64(11):1187–97.
60. Higgins JP, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *BMJ.* 2003;327(7414):557–60.
61. Sterne JA, Egger M, Moher D. Chapter 10: Addressing reporting biases. In: Higgins JPT, Green S, editors. *Conchrane handbook for systematic reviews of interventions*. Chichester: Wiley; 2008. p. 297–333.
62. Morales-Artacho AJ, García-Ramos A, Pérez-Castilla A, Padial P, Gomez AM, Peinado AM, et al. Muscle activation during power-oriented resistance training: continuous vs. cluster set configurations. *J Strength Cond Res.* 2018. <https://doi.org/10.1519/JSC.0000000000002811>.
63. Kurt C, Kafkas ME, Kurtdere A, Selalmaz O. Influence of traditional and cluster set plyometric warm-ups on reactive strength index and leg stiffness in male rugby players. *Isokinet Exerc Sci.* 2018;26(3):237–44.
64. Koefoed N, Lerche M, Jensen BK, Kjær PIA, Dam S, Horslev R, et al. Peak power output in loaded jump squat exercise is affected by set structure. *J Exerc Sci.* 2018;11(1):776–84.
65. Hansen KT, Cronin JB, Newton MJ. The effect of cluster loading on force, velocity, and power during ballistic jump squat training. *Int J Sports Physiol Perform.* 2011;6(4):455–68.
66. Cormie P, McCauley GO, McBride JM. Power versus strength-power jump squat training: influence on the load-power relationship. *Med Sci Sports Exerc.* 2007;39(6):996–1003.

67. Hardee JP, Lawrence MM, Zwetsloot KA, Triplett NT, Utter AC, McBride JM. Effect of cluster set configurations on power clean technique. *J Sports Sci*. 2013;31(5):488–96.
68. Wagle JP, Cunanan AJ, Carroll KM, Sams ML, Wetmore A, Bingham GE, et al. Accentuated eccentric loading and cluster set configurations in the back squat: a kinetic and kinematic analysis. *J Strength Cond Res*. 2018. <https://doi.org/10.1519/jsc.00000000000002677>.
69. Wetmore A, Wagle JP, Sams ML, Taber C, DeWeese BH, Sato K, et al. Cluster set loading in the back squat: kinetic and kinematic implications. *J Strength Cond Res*. 2019. <https://doi.org/10.1519/JSC.00000000000002972>.
70. Oliver JM, Kreutzer A, Jenke S, Phillips MD, Mitchell JB, Jones MT. Acute response to cluster sets in trained and untrained men. *Eur J Appl Physiol*. 2015;115(11):2383–93.
71. Davies TB, Halaki M, Orr R, Helms ER, Hackett DA. Changes in bench press velocity and power after 8 weeks of high-load cluster- or traditional-set structures. *J Strength Cond Res*. 2019. <https://doi.org/10.1519/jsc.00000000000003166>.
72. Ho IMK, Luk JC, Ngo JK, Wong DP. Effects of different intraset rest durations on lifting performance and self-perceived exertion during bench press exercise. *J Strength Cond Res*. 2019. <https://doi.org/10.1519/jsc.00000000000003101>.
73. Iglesias-Soler E, Carballeira E, Sanchez-Otero T, Mayo X, Fernandez-Del-Olmo M. Performance of maximum number of repetitions with cluster-set configuration. *Int J Sports Physiol Perform*. 2014;9(4):637–42.
74. Wagle JP, Taber CB, Carroll KM, Cunanan AJ, Sams ML, Wetmore A, et al. Repetition-to-repetition differences using cluster and accentuated eccentric loading in the back squat. *Sports (Basel)*. 2018;6(3):59.
75. Girman JC, Jones MT, Matthews TD, Wood RJ. Acute effects of a cluster-set protocol on hormonal, metabolic and performance measures in resistance-trained males. *Eur J Sport Sci*. 2014;14(2):151–9.
76. Moreno SD, Brown LE, Coburn JW, Judelson DA. Effect of cluster sets on plyometric jump power. *J Strength Cond Res*. 2014;28(9):2424–8.
77. Cuevas-Aburto J, Jukic I, Chiroso-Ríos LJ, González Hernández J, Janicijevic D, Barboza-González P, et al. Effect of traditional, cluster, and rest redistribution set configurations on neuromuscular and perceptual responses during strength-oriented resistance training. *J Strength Cond Res*. 2020. <https://doi.org/10.1519/JSC.00000000000003658>.
78. Cuevas-Aburto J, Jukic I, González Hernández J, Janicijevic D, Barboza-González P, Chiroso L, et al. Effect of resistance training programs differing in the set configuration on maximal strength and explosive actions performance. *Int J Sports Physiol Perform*. 2020. [published ahead of print].
79. Sanchez-Medina L, González-Badillo JJ. Velocity loss as an indicator of neuromuscular fatigue during resistance training. *Med Sci Sports Exerc*. 2011;43(9):1725–34.
80. Jimenez-Reyes P, Pareja-Blanco F, Cuadrado-Peñafiel V, Morcillo J, Párraga J, González-Badillo J. Mechanical, metabolic and perceptual response during sprint training. *Int J Sports Med*. 2016;37(10):807–12.
81. Morcillo JA, Jiménez-Reyes P, Cuadrado-Peñafiel V, Lozano E, Ortega-Becerra M, Párraga J. Relationships between repeated sprint ability, mechanical parameters, and blood metabolites in professional soccer players. *J Strength Cond Res*. 2015;29(6):1673–82.
82. Lagally KM, Robertson RJ, Gallagher KI, Goss FL, Jakicic JM, Lephart SM, et al. Perceived exertion, electromyography, and blood lactate during acute bouts of resistance exercise. *Med Sci Sports Exerc*. 2002;34(3):552–9.
83. Oliver JM, Kreutzer A, Jenke SC, Phillips MD, Mitchell JB, Jones MT. Velocity drives greater power observed during back squat using cluster sets. *J Strength Cond Res*. 2016;30(1):235–43.
84. Stone JD, King AC, Goto S, Mata JD, Hannon J, Garrison JC, et al. Joint-level analyses of the back squat with and without intra-set rest. *Int J Sports Physiol Perform*. 2019;14(5):583–9.
85. Singh F, Foster C, Tod D, McGuigan MR. Monitoring different types of resistance training using session rating of perceived exertion. *Int J Sports Physiol Perform*. 2007;2(1):34–45.
86. Vernon A, Joyce C, Banyard HG. Readiness to train: Return to baseline strength and velocity following strength or power training. *Int J Sports Sci Coach*. 2020. <https://doi.org/10.1177/1747954119900120>.
87. Weakley J, Ramirez-Lopez C, McLaren S, Dalton-Barron N, Weaving D, Jones B, et al. The effects of 10%, 20%, and 30% velocity loss thresholds on kinetic, kinematic, and repetition characteristics during the barbell back squat. *Int J Sports Physiol Perform*. 2019. <https://doi.org/10.1123/ijspp.2018-1008>.
88. González-Badillo JJ, Rodríguez-Rosell D, Sánchez-Medina L, Ribas J, López-López C, Mora-Custodio R, et al. Short-term recovery following resistance exercise leading or not to failure. *Int J Sports Med*. 2016;37(04):295–304.
89. Pareja-Blanco F, Rodríguez-Rosell D, Aagaard P, Sánchez-Medina L, Ribas-Serna J, Mora-Custodio R, et al. Time course of recovery from resistance exercise with different set configurations. *J Strength Cond Res*. 2018. <https://doi.org/10.1519/JSC.00000000000002756>.
90. Pareja-Blanco F, Villalba-Fernández A, Cornejo-Daza PJ, Sánchez-Valdepeñas J, González-Badillo JJ. Time course of recovery following resistance exercise with different loading magnitudes and velocity loss in the set. *Sports*. 2019;7(3):59.
91. González-Badillo JJ, Pareja-Blanco F, Rodríguez-Rosell D, Abad-Herencia JL, del Ojo-López JJ, Sánchez-Medina L. Effects of velocity-based resistance training on young soccer players of different ages. *J Strength Cond Res*. 2015;29(5):1329–38.
92. Pareja-Blanco F, Sánchez-Medina L, Suárez-Arrones L, González-Badillo JJ. Effects of velocity loss during resistance training on performance in professional soccer players. *Int J Sports Physiol*. 2017;12(4):512–9.
93. Pareja-Blanco F, Rodríguez-Rosell D, Sánchez-Medina L, Sanchis-Moysi J, Dorado C, Mora-Custodio R, et al. Effects of velocity loss during resistance training on athletic performance, strength gains and muscle adaptations. *Scand J Sci Med Sports*. 2017;27(7):724–35.
94. Schoenfeld BJ, Grgic J, Ogborn D, Krieger JW. Strength and hypertrophy adaptations between low vs. high-load resistance training: a systematic review and meta-analysis. *J Strength Cond Res*. 2017;31(12):3508–23.
95. Ralston GW, Kilgore L, Wyatt FB, Baker JS. The effect of weekly set volume on strength gain: a meta-analysis. *Sports Med*. 2017;47(12):2585–601.
96. Mangine GT, Hoffman JR, Gonzalez AM, Townsend JR, Wells AJ, Jajtner AR, et al. The effect of training volume and intensity on improvements in muscular strength and size in resistance-trained men. *Physiol Rep*. 2015;3(8):e12472.
97. Denton J, Cronin JB. Kinematic, kinetic, and blood lactate profiles of continuous and intraset rest loading schemes. *J Strength Cond Res*. 2006;20(3):528–34.
98. Joy J, Oliver J, McCleary S, Lowery R, Wilson J. Power output and electromyography activity of the back squat exercise with cluster sets. *J Sports Sci*. 2013;1:37–45.

Affiliations

Ivan Jukic¹  · Amador García Ramos^{2,3}  · Eric R. Helms¹ · Michael R. McGuigan¹  · James J. Tufano⁴ 

✉ Ivan Jukic
ivan.jukic@aut.ac.nz

¹ Sport Performance Research Institute New Zealand (SPRINZ), Auckland University of Technology, Auckland, New Zealand

² Department of Physical Education and Sport, Faculty of Sport Sciences, University of Granada, Granada, Spain

³ Department of Sports Sciences and Physical Conditioning, Faculty of Education, Universidad Católica de la Santísima Concepción, Concepción, Chile

⁴ Department of Physiology and Biochemistry, Faculty of Physical Education and Sport, Charles University, Prague, Czech Republic